Variable plots

- First sample: simulated signal 23k events → slide 3
  - Correlation between variables → slide 9

- Recovered data and code files → start in slide 20

Process: $pp \rightarrow pX(\rightarrow \tau\tau)p$;

Protons are tagged by PPS, while the decay products of the X system are detected by the CMS central detector.

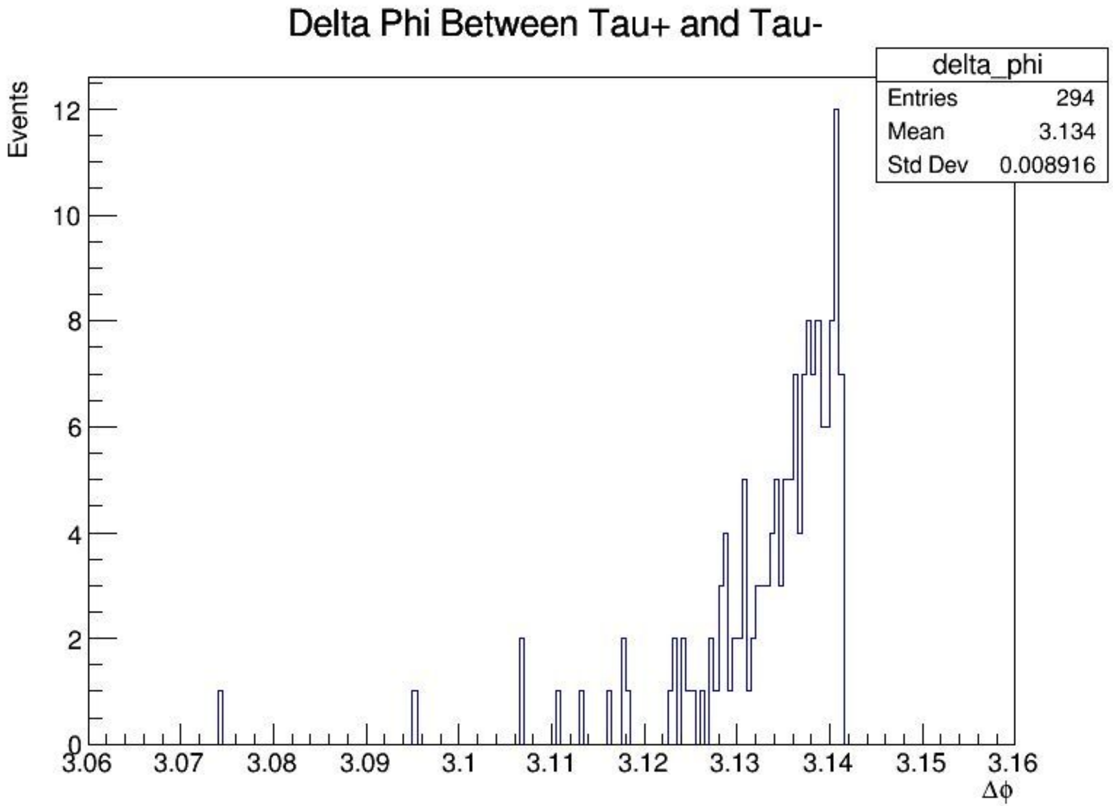
Decay mode in study:

- Fully hadronic decay mode; Both τs decay hadronically

Fase0 code 23k -> 294 events
127 for no pileup protons

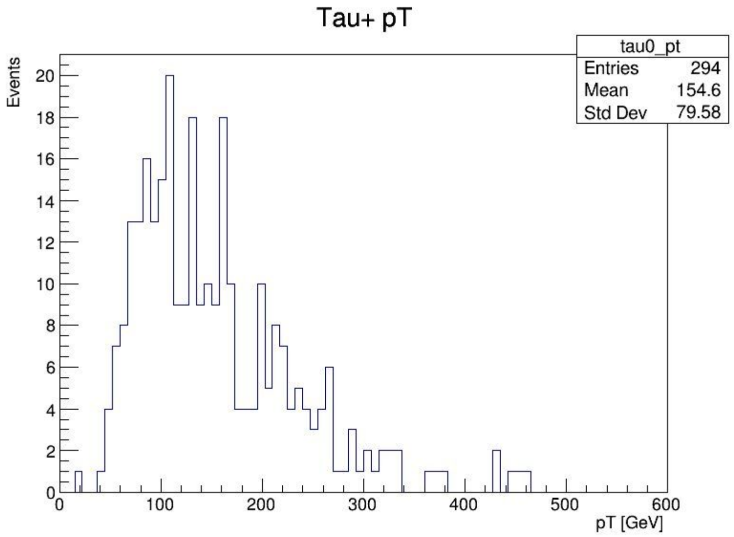
- Monte Carlo generated signal (23k events before processing);
- Events selected : Where trigger `HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg` is applied
- Protons are selected by checking if there is one on each arm. If there is more than one, the one with largest ξ is selected.
- proton selection was fixed
- Inv mass calculation for tau pair was changed

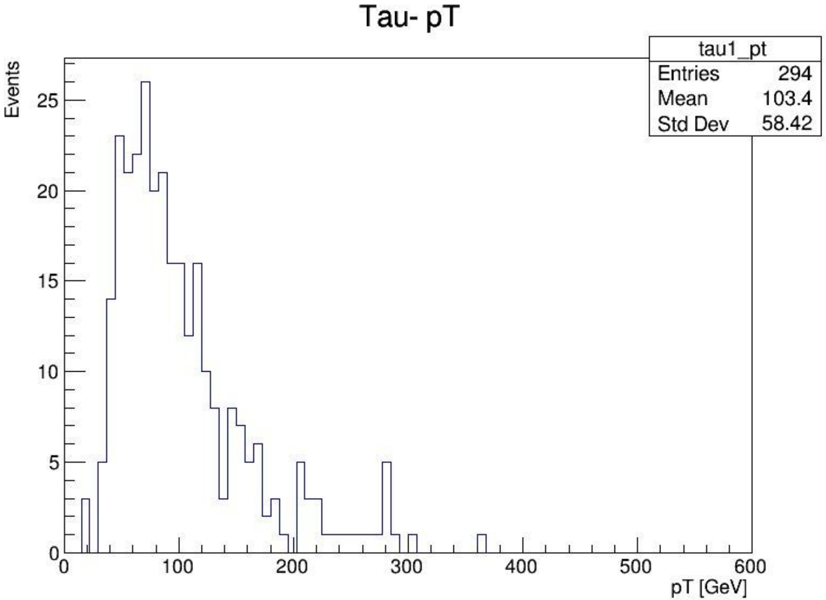
Δϕ



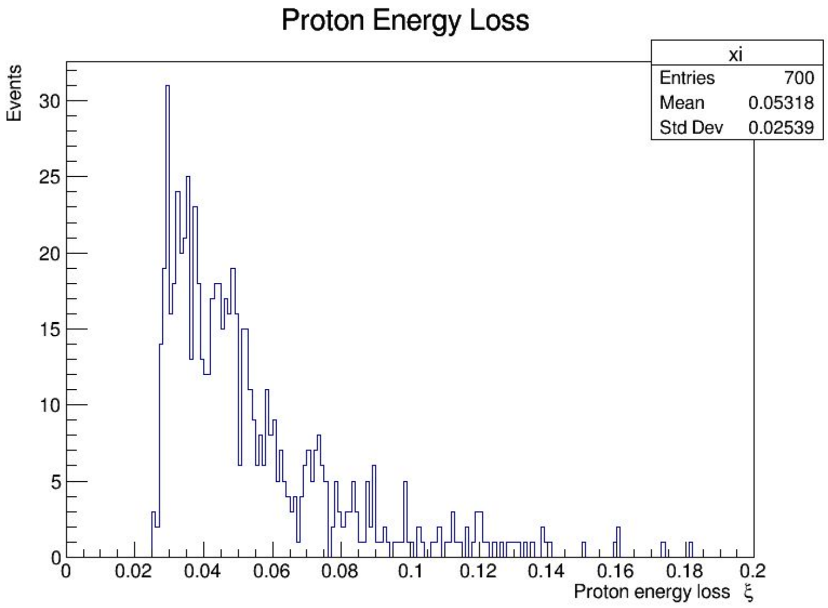
Delta Phi Between Tau+ and Tau-
delta_phi
Entries 294
Mean 3.134
Std Dev 0.008916
Events
0
2
4
6
8
10
12
3.06
3.07
3.08
3.09
3.1
3.11
3.12
3.13
3.14
3.15
3.16
Δϕ

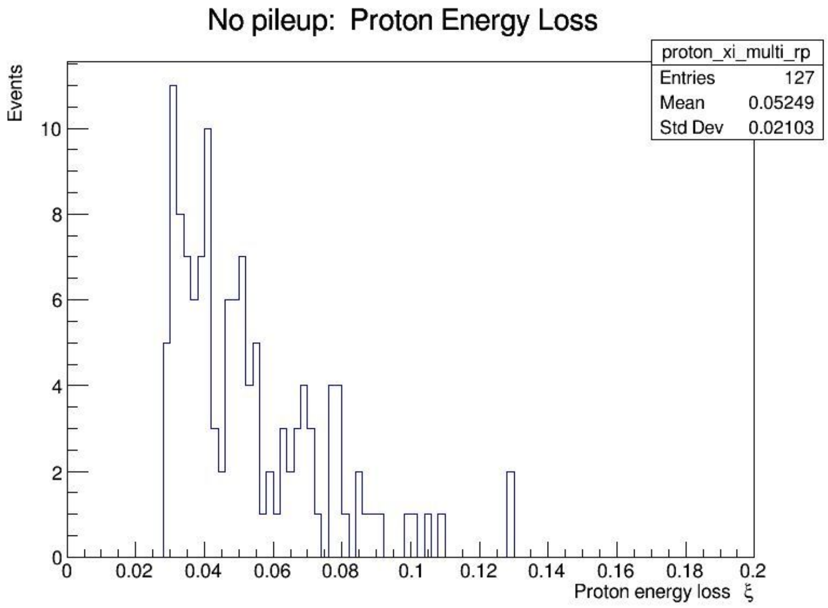
Pseudorapidity, η



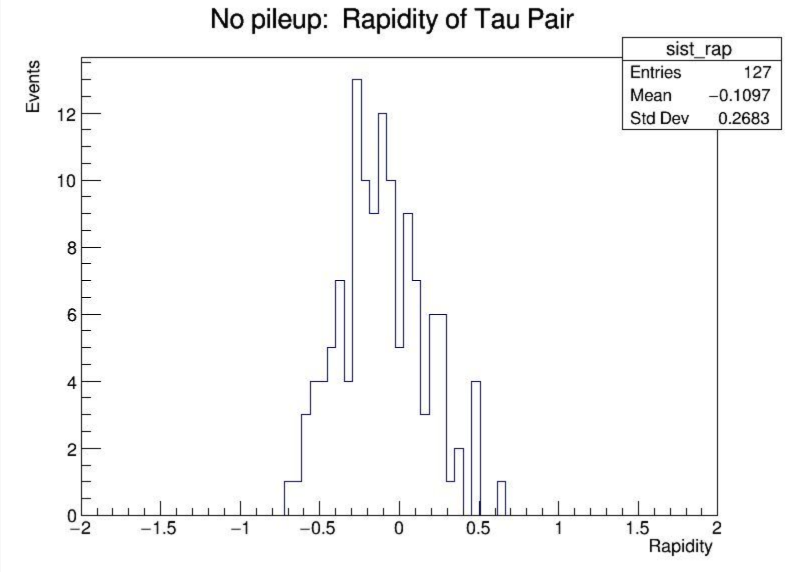


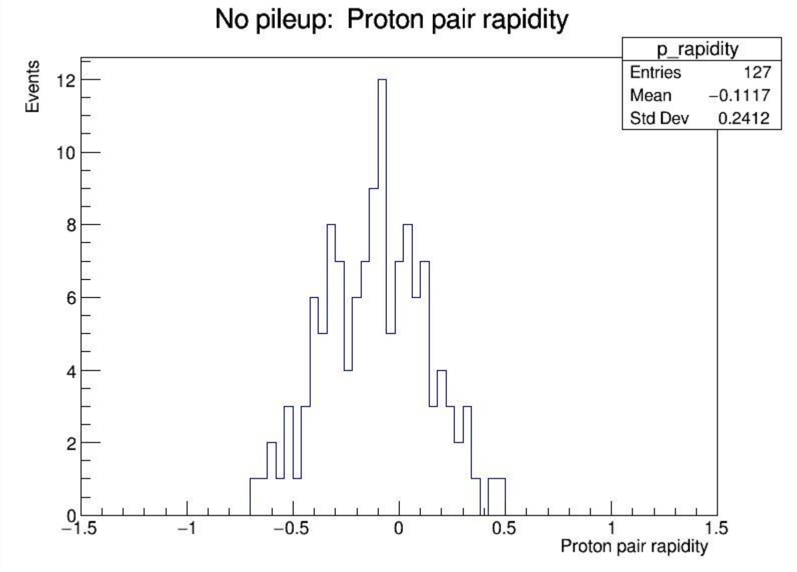
Proton energy Loss, ξ





Plots



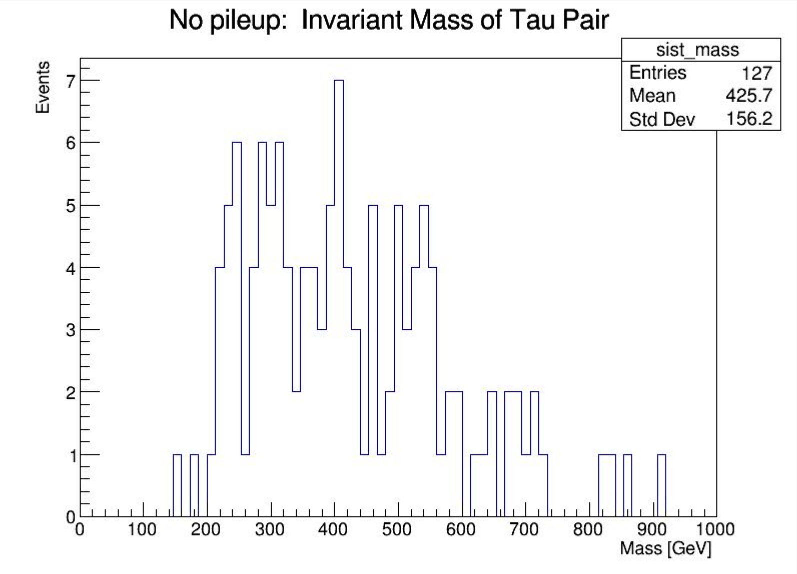


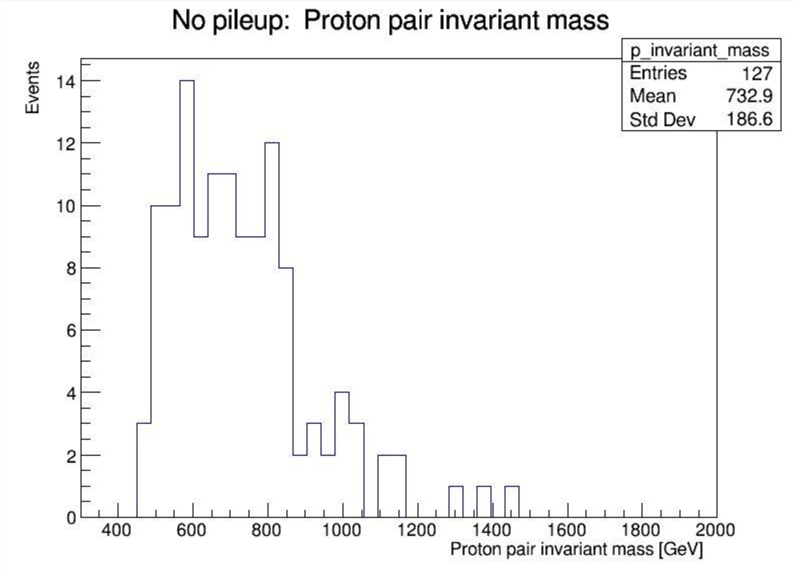
$$Y_X = \frac{1}{2}(\frac{E+p_z}{E-p_z})$$

$$Y_X = \frac{1}{2}log(\xi_1/\xi_2)$$

- E is total energy of tau pair
- P_z is total longitudinal moment of tau pair system

Plots





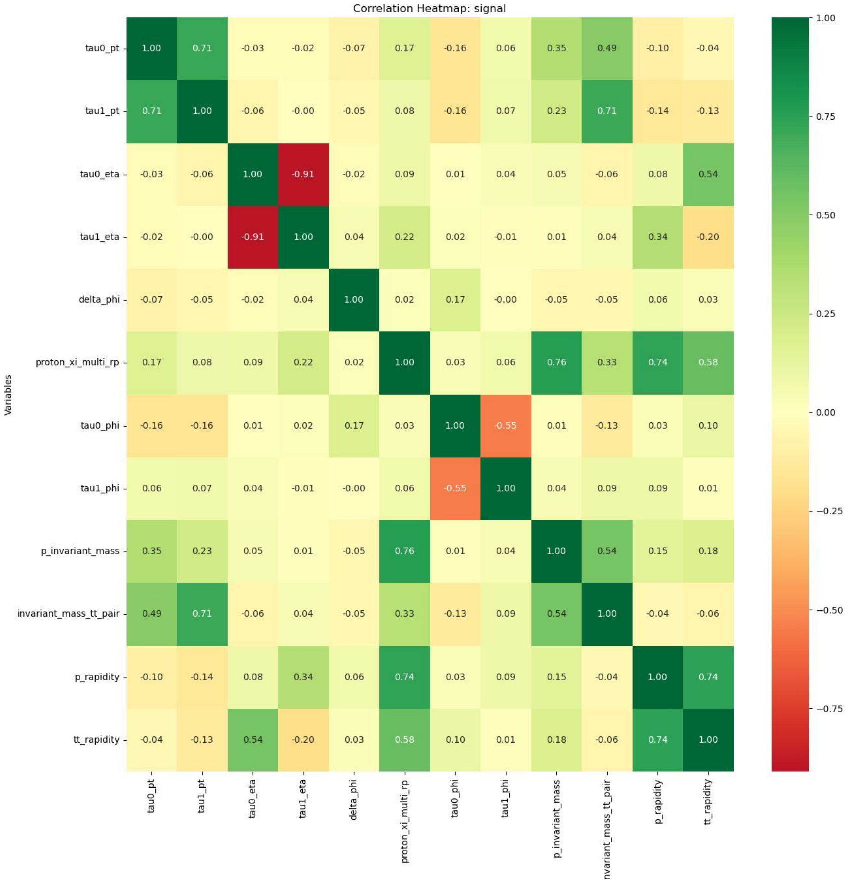
Changed to formula that uses

`TLorentzVector`

$$M_X^2 = s\xi_1\xi_2$$

Correlation between variables

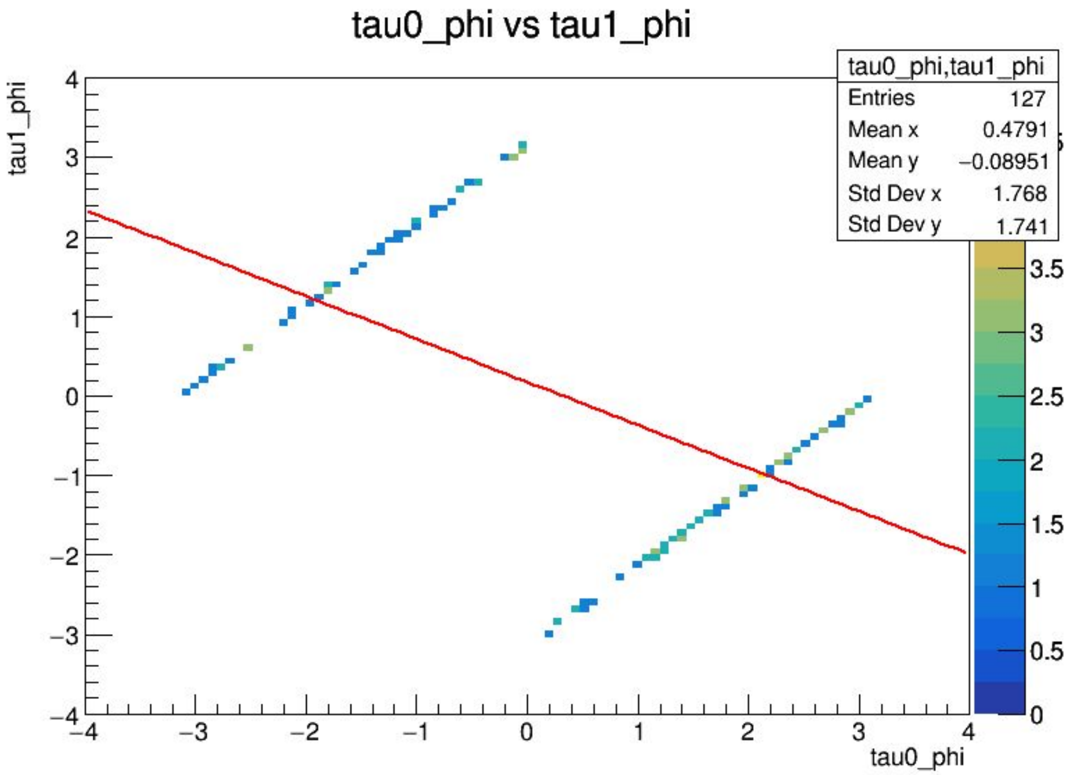
- tau0_phi,tau1_phi
- tau0_eta,tau1_eta
- tau0_pt,tau1_pt
- sist_mass,tau0_pt
- sist_mass,tau1_pt
- Sist_rap,tau0_eta
- Tt_rapidity,tau0_eta
- p_invariant_mass,invariant_mass_tt_pair
- p_rapidity,tt_rapidity



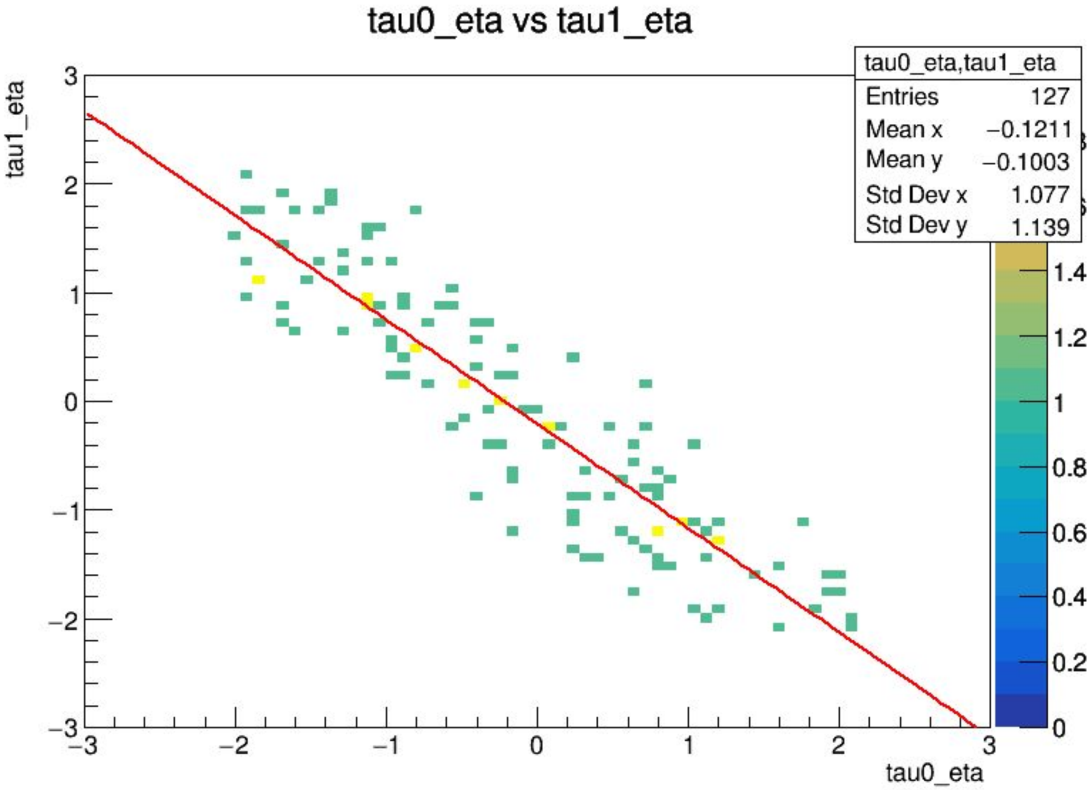
Expected:

- Strong correlation between tau0 and tau1 phi
- Negative correlation between tau0 and tau1 eta
- Strong correlation between invariant mass of tau pair and tau0 and tau1 pt
- Some correlation between tau pair rapidity and tau0 eta

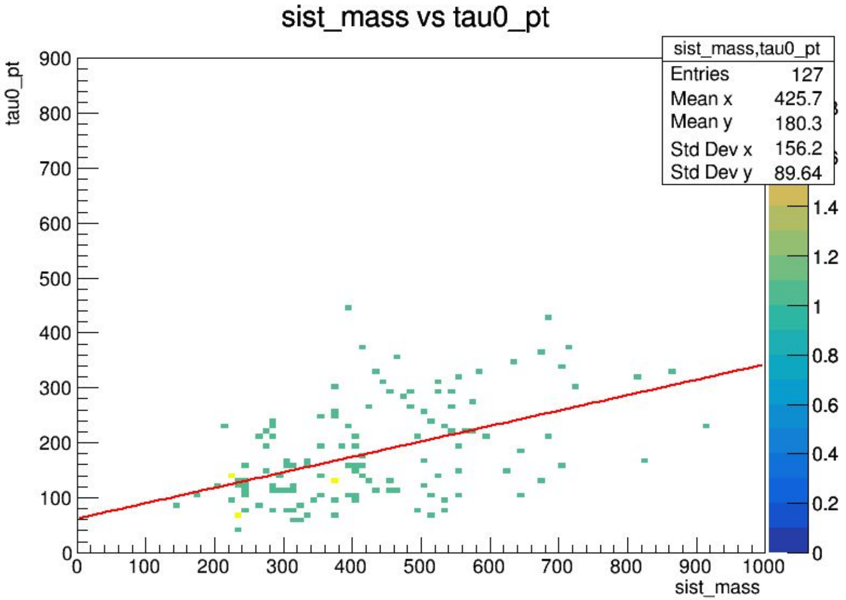
Tau0 and Tau1 phi

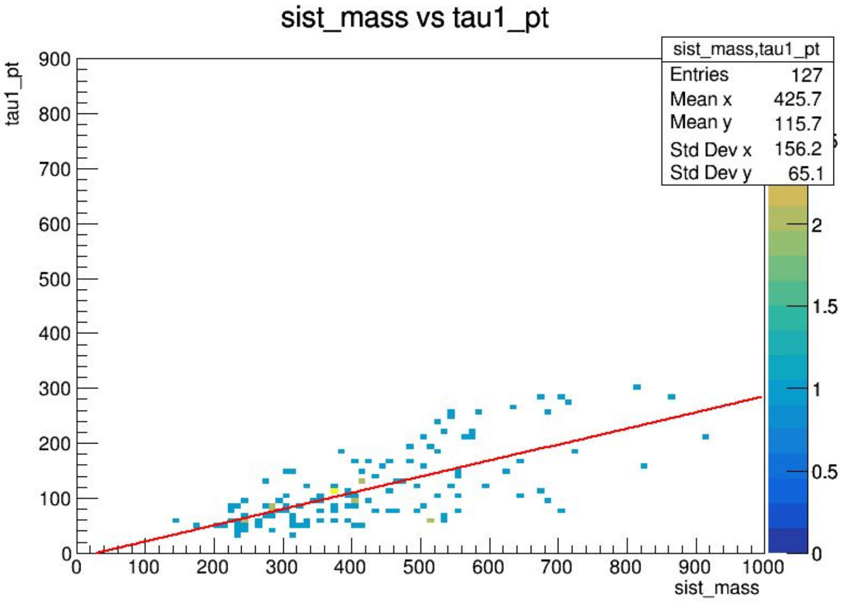


Tau0 and Tau1 eta

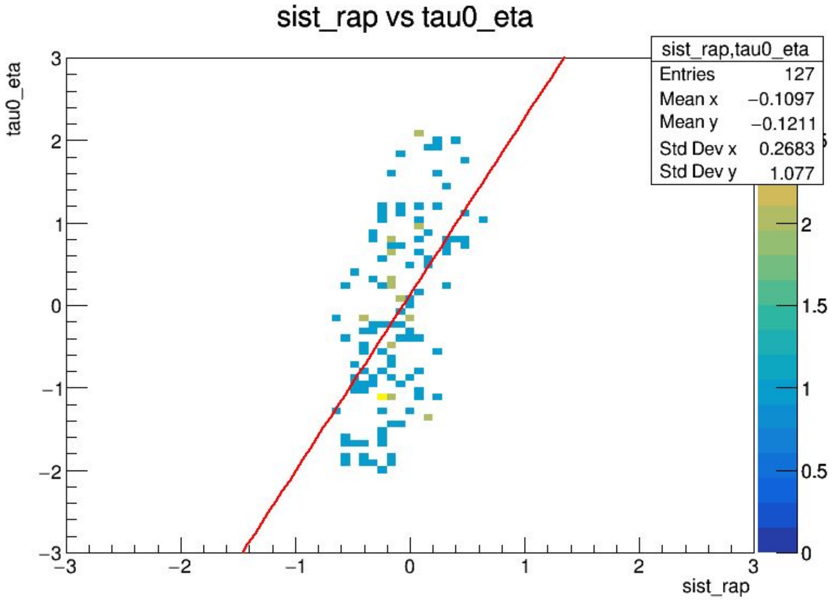


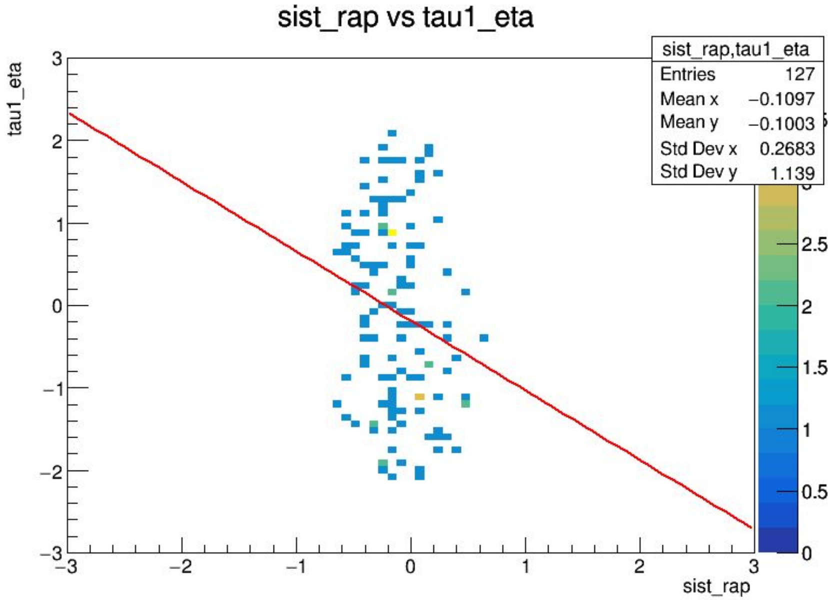
System mass, tau0 pT and tau1 pT



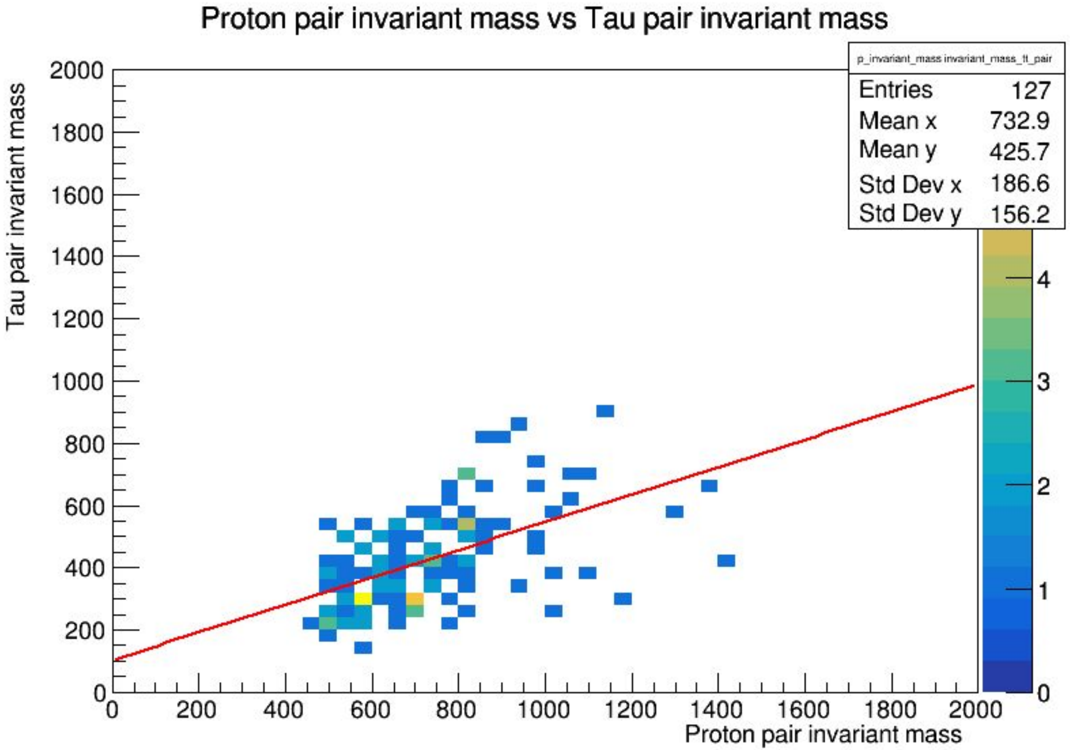


System rapidity, tau0 eta

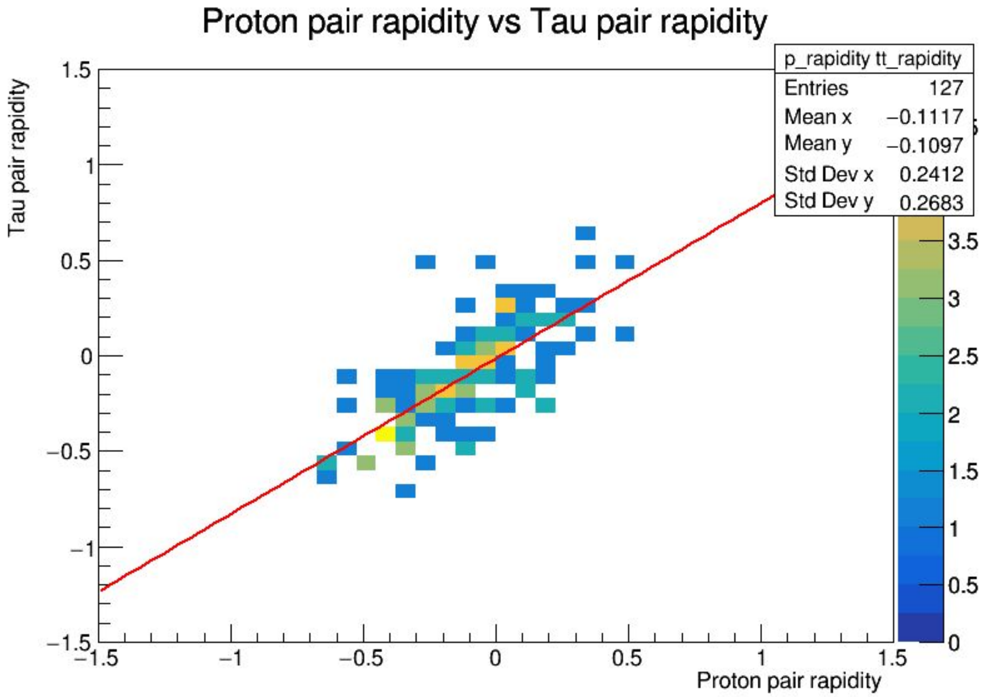




Invariant mas of tau pair, protons



Proton rapidity, tau pair rapidity

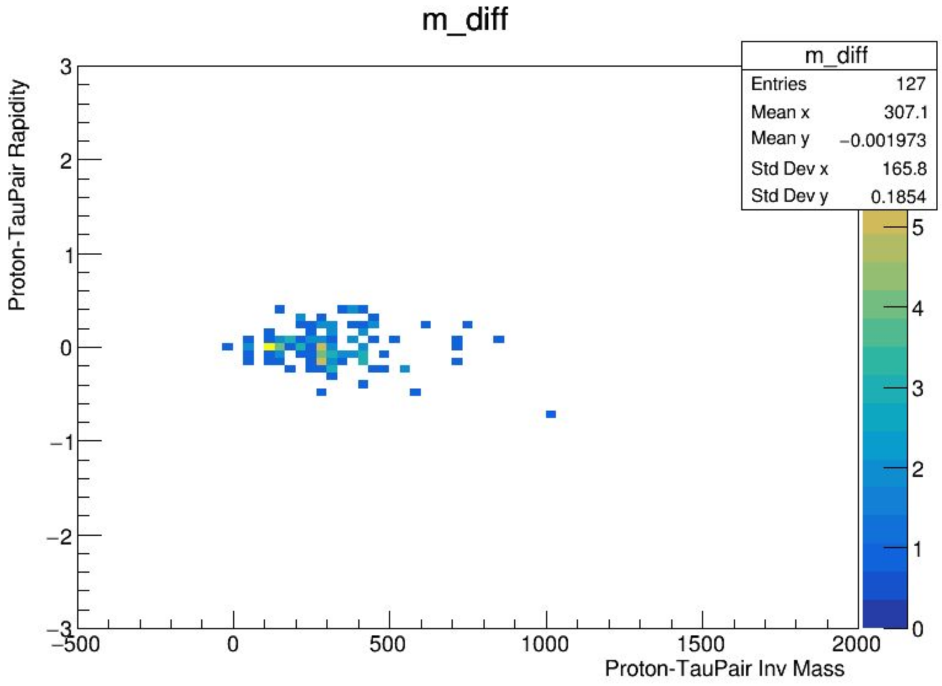


Proton-Tau pair invariant mass, Rapidity

Proton inv mass - tau pair inv mass

Vs

Proton rapidity - tau rapidity



		fase0		fase0		fase1	
Column 1	condition applied	with pileups	% of original data	# no pileups	% of original data	with pile	no pileu
	no cuts	~~23059~~	100	16837	100		
using HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1	trigger only	910	3.946398369	664	3.943695433		
	trigger and 2 or more taus	700	3.035691053	508	3.017164578		
	trigger and 1 proton on each arm	381	1.652283273	168	0.9978024589		
if(tree->GetLeaf("Tau_pt")->GetLen() < 2) continue;	2 taus or more	3963	17.18634806	2889	17.15863871		
	2 taus and one proton on each arm	1602	6.947395811	609	3.617033913		
if (arm == 0 && xi > proton_xi1) { // } else if (arm == 1 && xi > proton_xi2) {	1 proton on each arm	8874	38.48388915	3113	18.48904199		
	all 3	294	1.274990242	127	0.7542911445	55	31

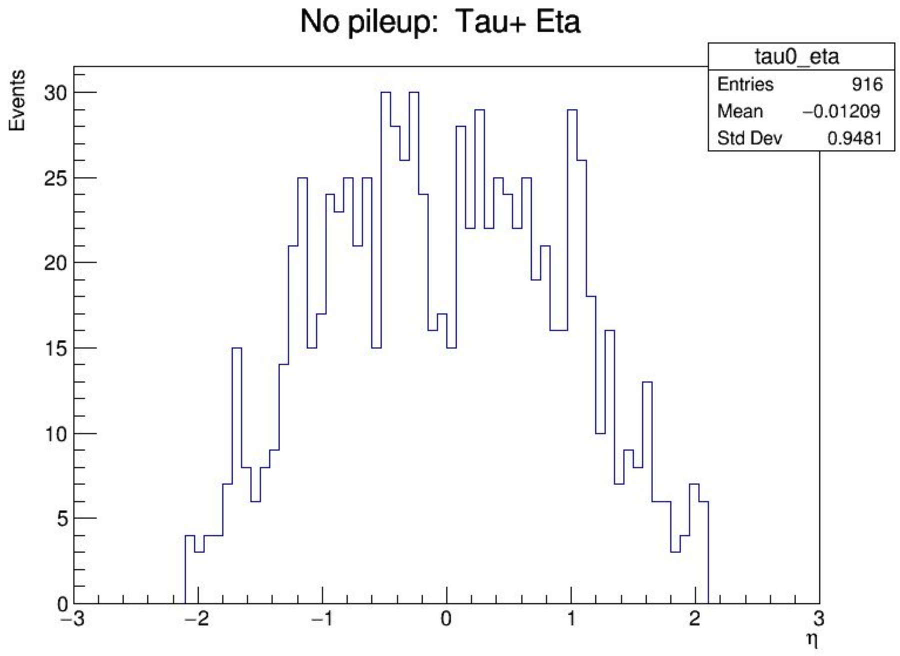
Tau cuts

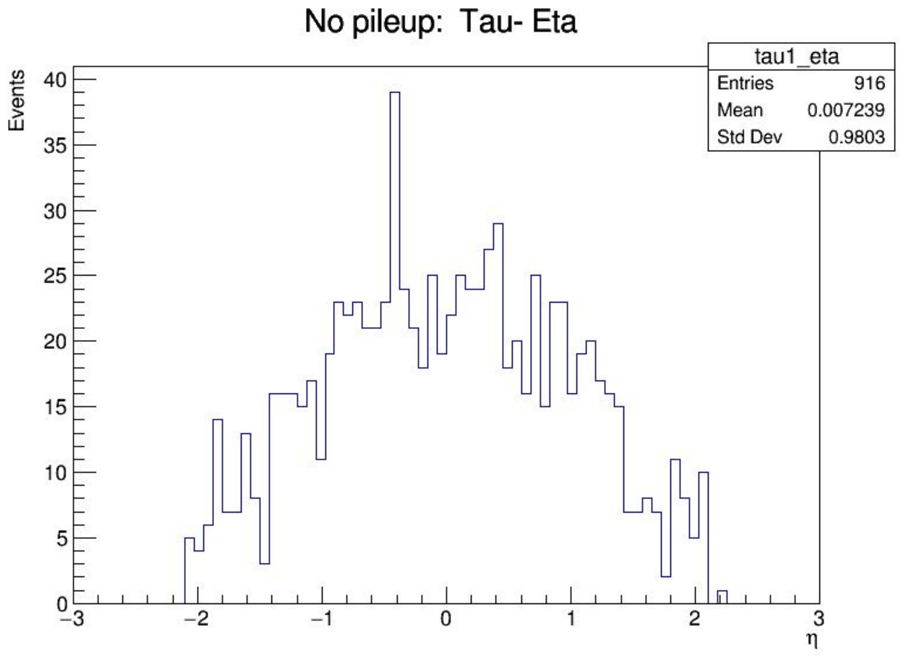
`Tau_idDeepTau2017v2p1VSjet`	6 → Tight
`Tau_idDeepTau2017v2p1VSe`	3 → VLoose
`Tau_idDeepTau2017v2p1VSmu`	1 → VVVLoose
`DeltaR`	>0.4
`tau0.Eta and tau1.Eta`	<2.4
`Tau0_pt and tau1_pt`	>100
`Opposite charge`	✔

New data and code

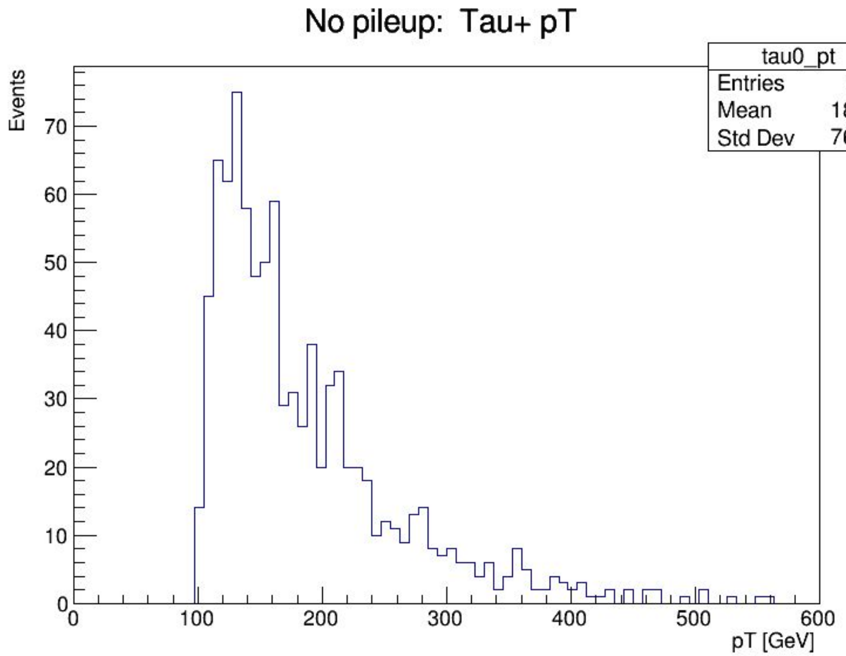
Variable name	Description

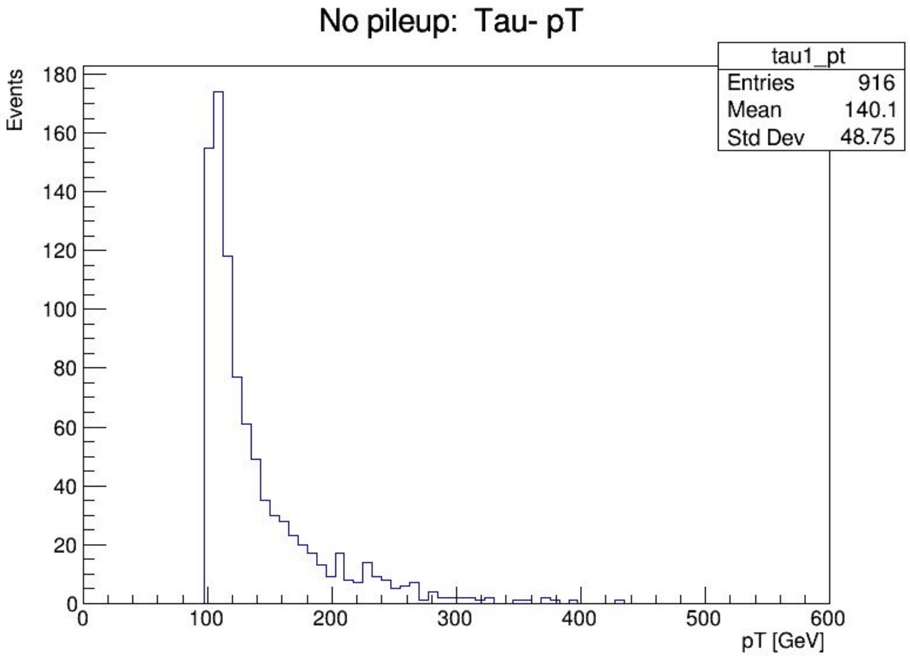
Plots full background with signal-like selection (opposite sign taus)



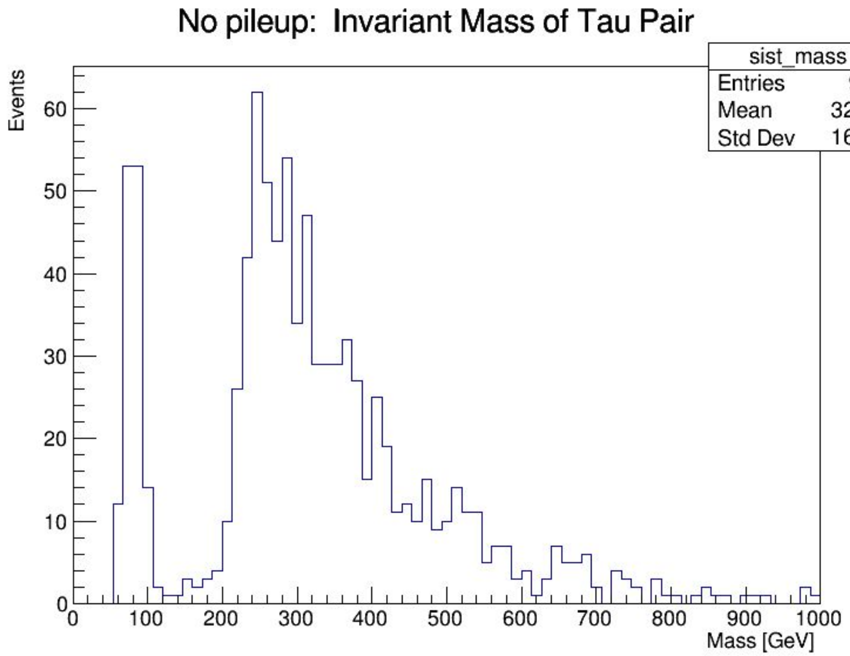


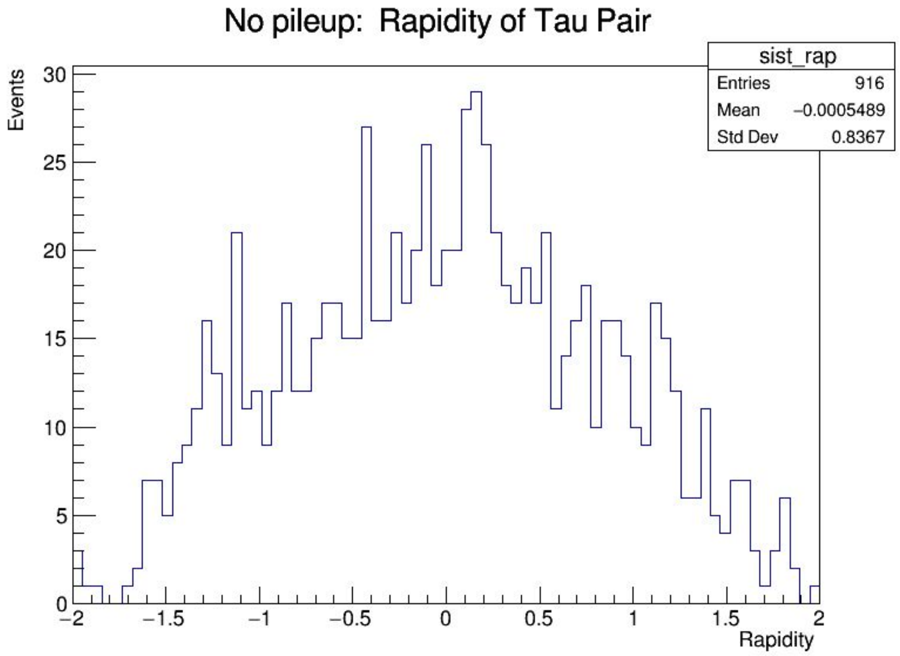
Plots full background with signal-like selection (opposite sign taus)



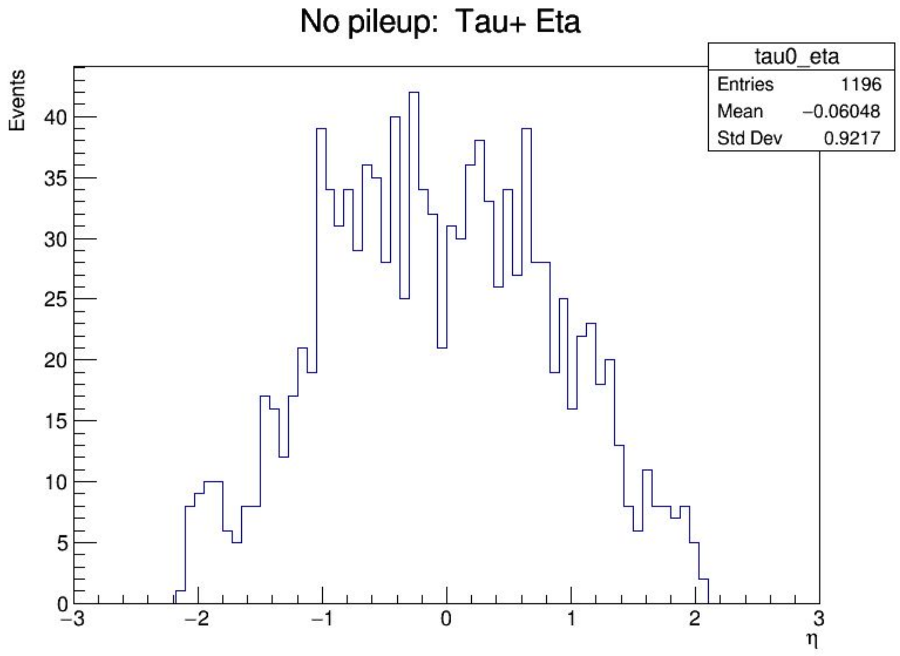


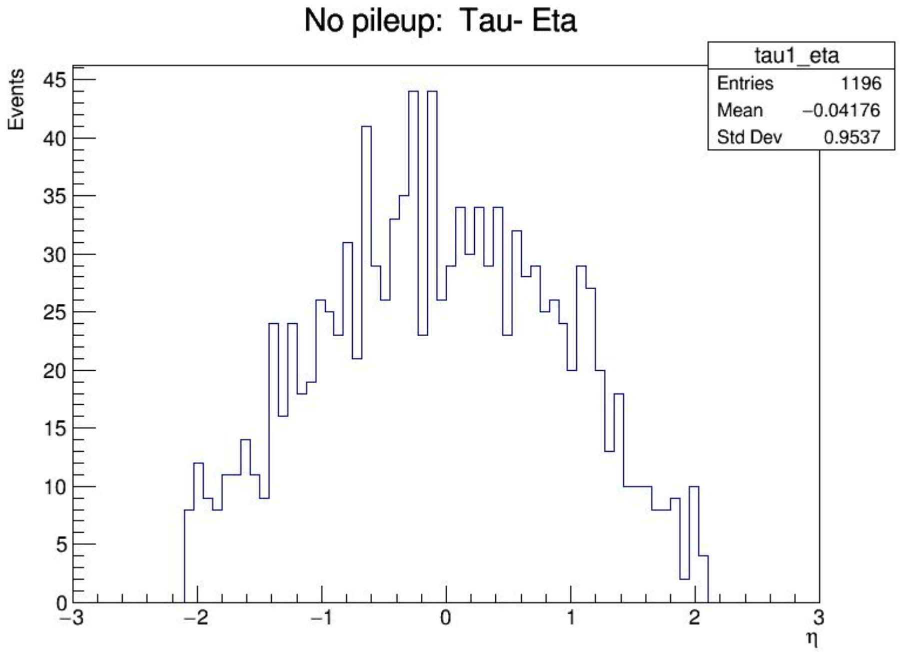
Plots full background with signal-like selection (opposite sign taus)



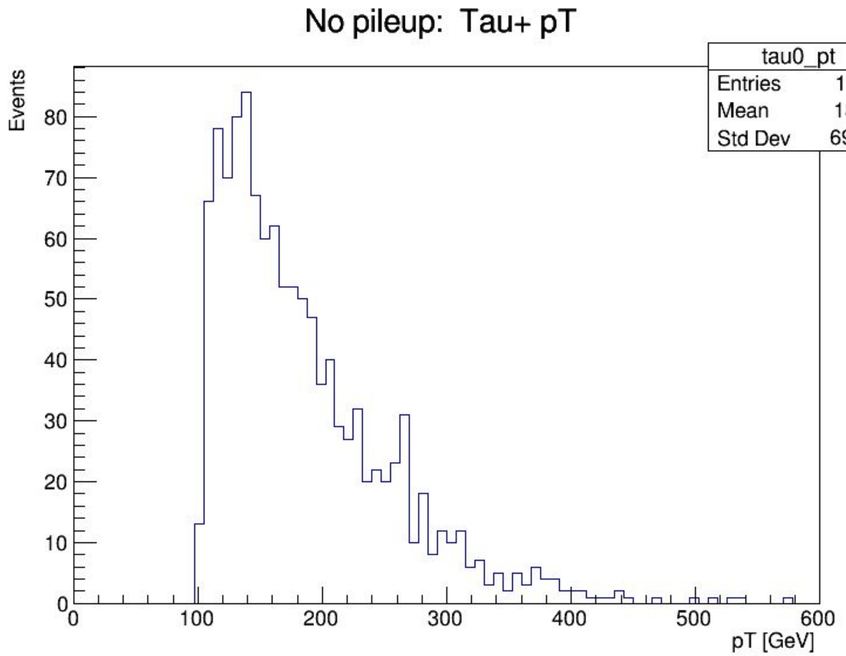


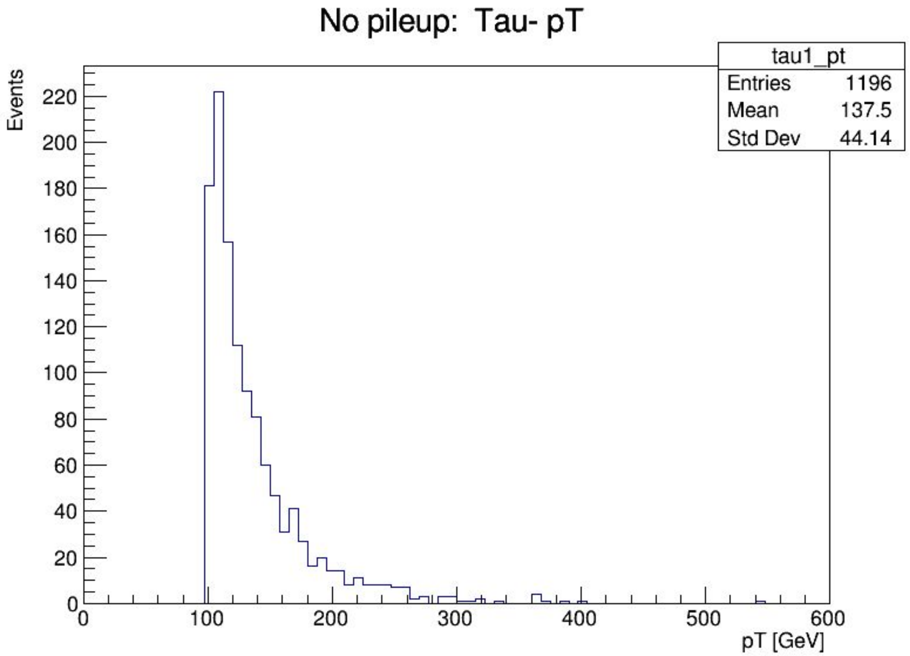
Plots fase1 files for the sum of all backgrounds, including pileup protons



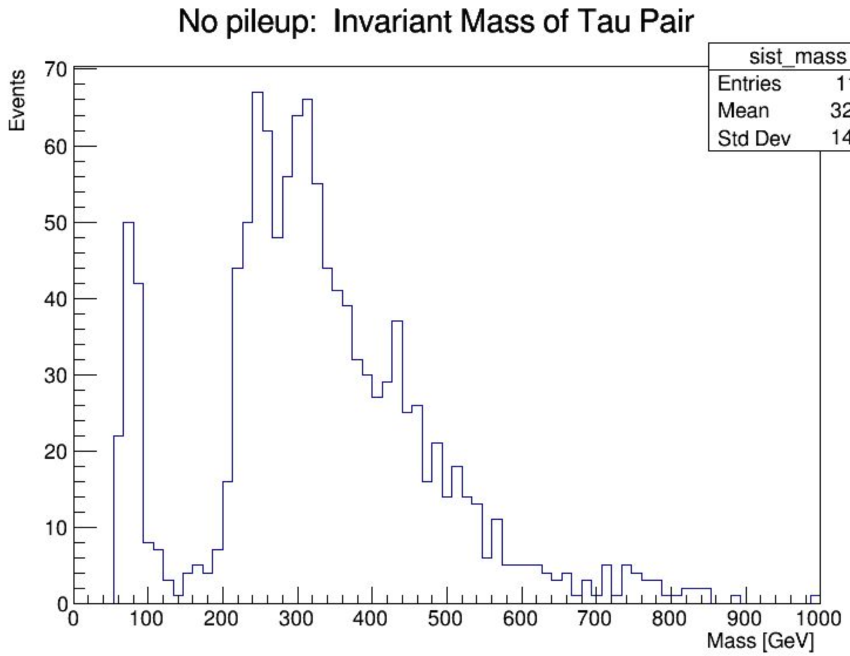


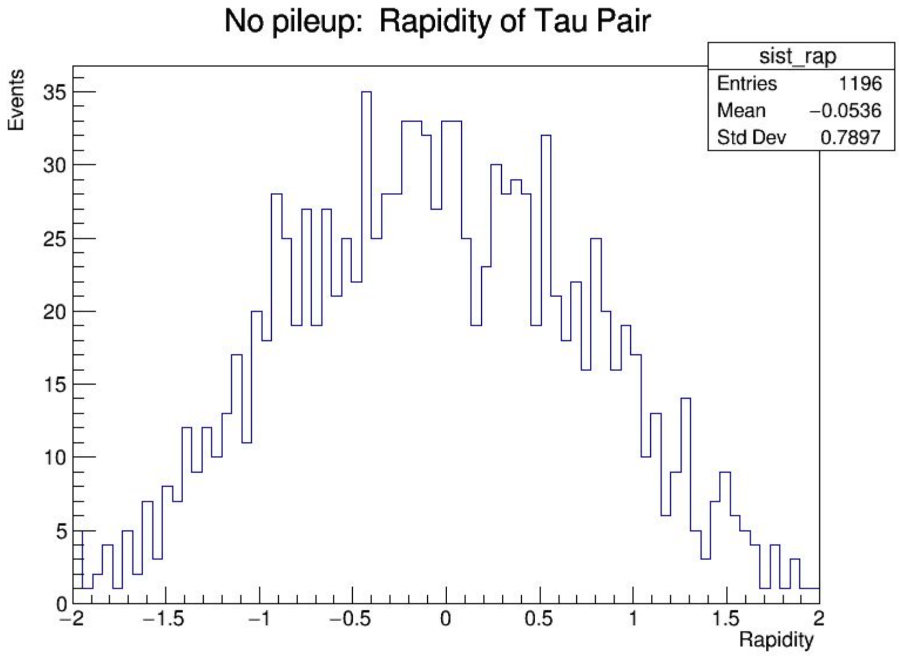
Plots fase1 files for the sum of all backgrounds, including pileup protons





Plots fase1 files for the sum of all backgrounds, including pileup protons



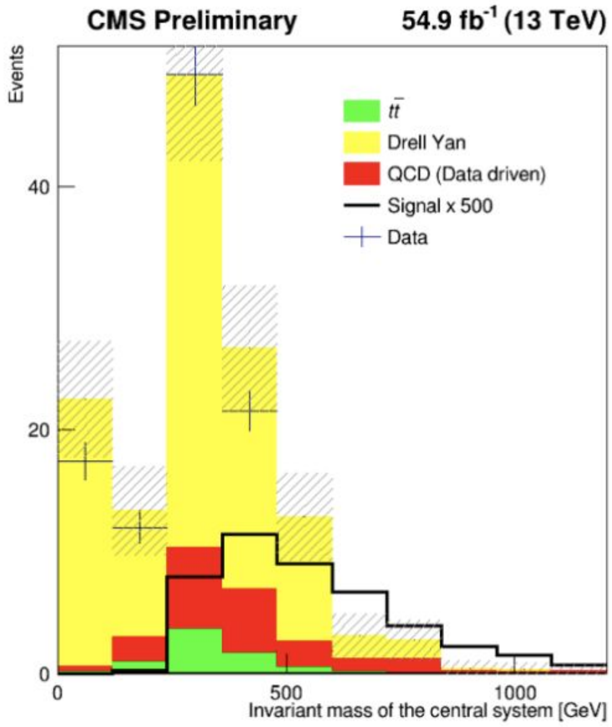


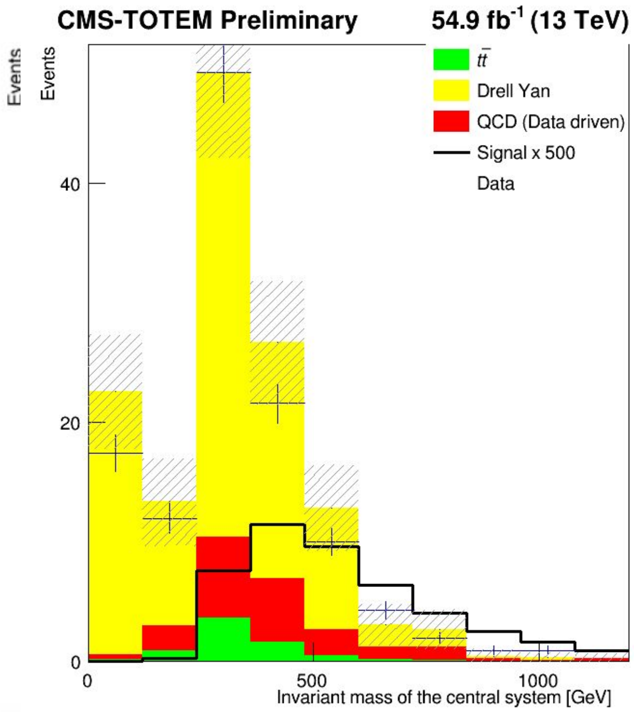
Input distributions to the BDT discriminator

- Invariant mass of the central system
  → The total mass of all particles in the central detector, calculated from their 4-momenta.
- Rapidity of the central system
  → Describes how forward or backward the central system is moving along the beam axis.
- Mass difference
  → Difference between the central system mass and the mass reconstructed using the forward protons (CMS - PPS).
- MET (Missing Transverse Energy)
  → Represents undetected particles.
- Transverse momentum of the central system
  → Momentum of the central system perpendicular to the beam axis.
- Rapidity matching
  → Comparison of rapidity measured centrally..
- Transverse momentum of hadronic tau
  → Momentum perpendicular to the beam of the hadronically decaying tau lepton.
- Acoplanarity of central system
  → Measures how "non-back-to-back" the two leading tau candidates are; zero means perfect balance.

Invariant mass of the central system

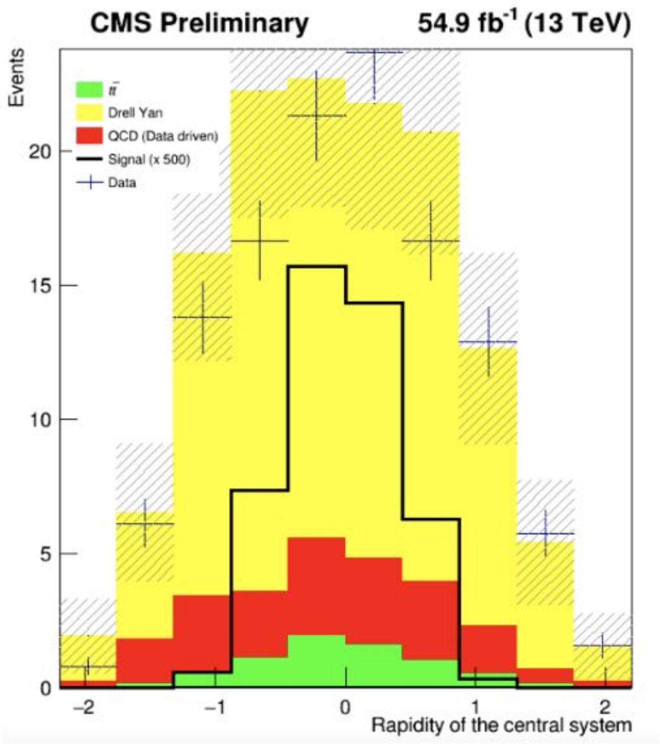
Original - left vs New - right

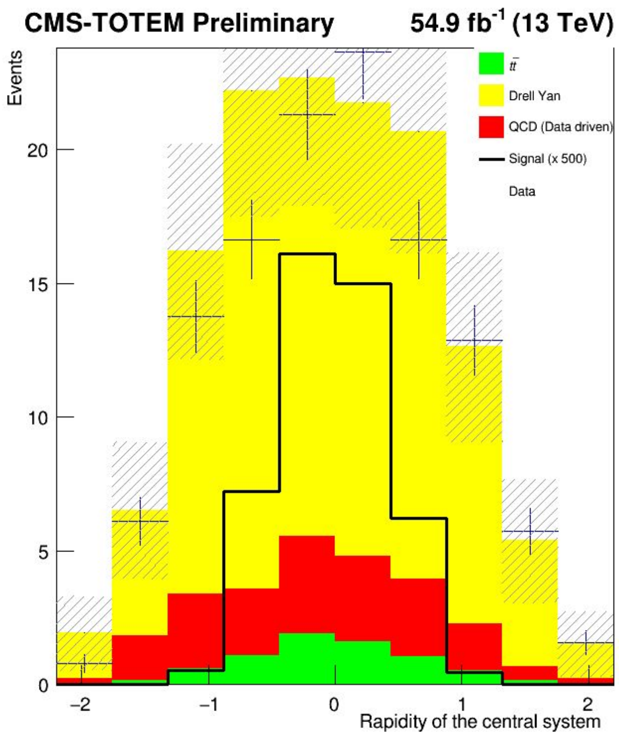




Rapidity of the central system

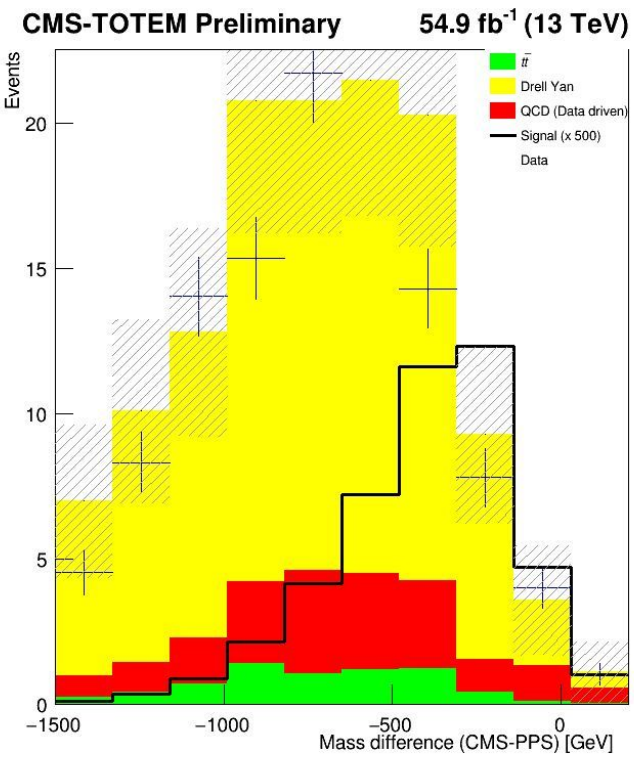
Original - left vs New - right

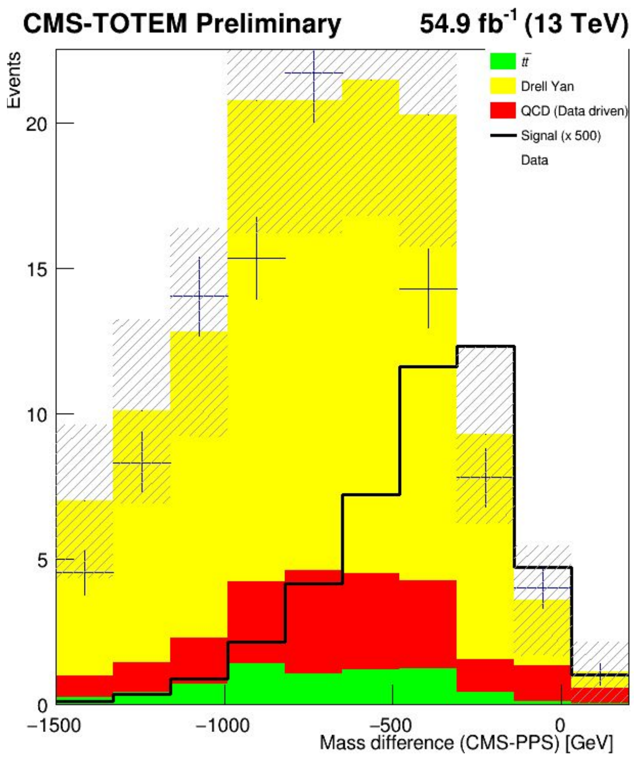




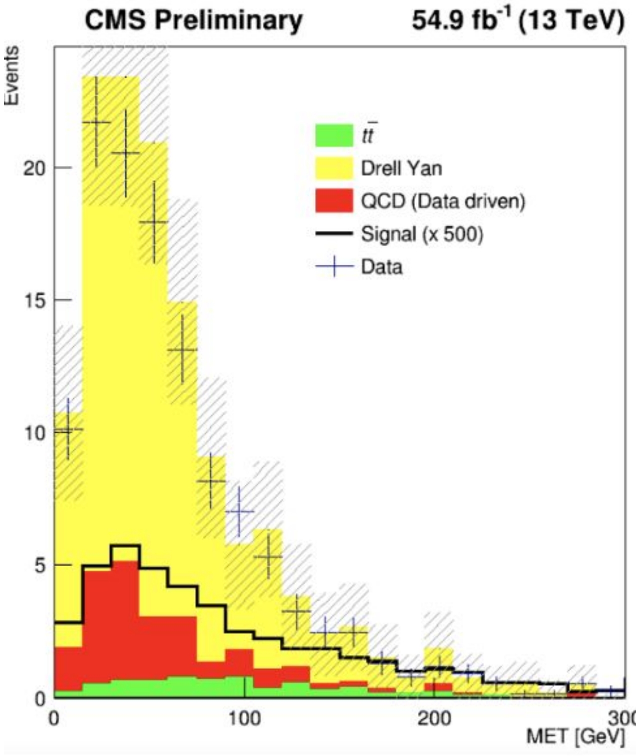
Mass Difference

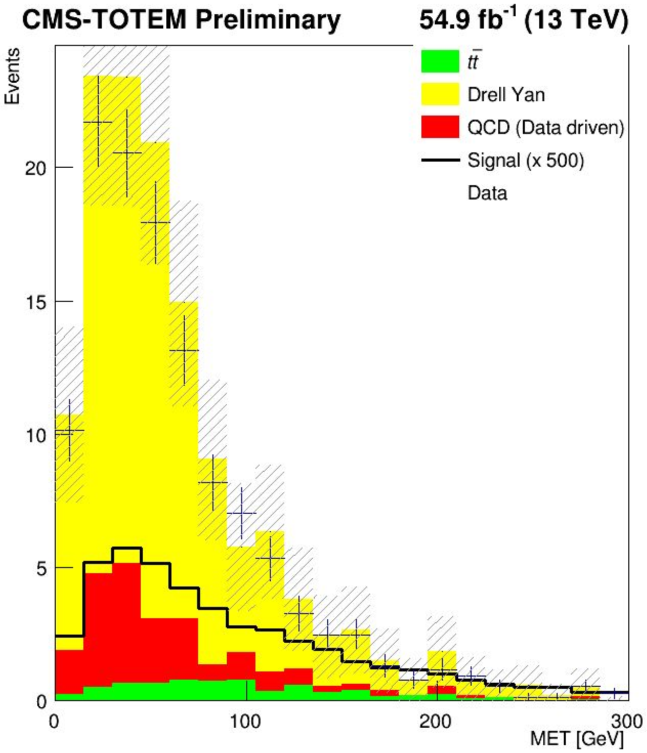
Original - left vs New - right





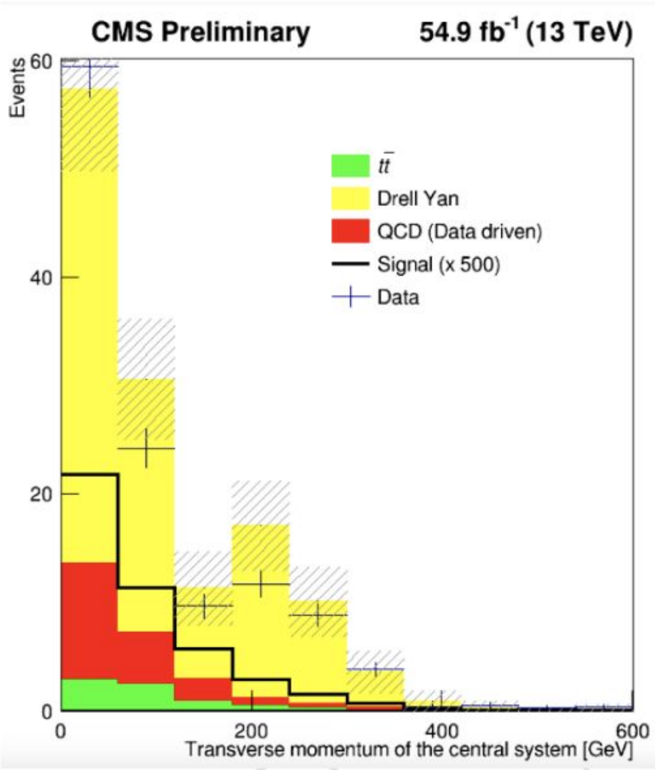
MET
Original - left vs New - right

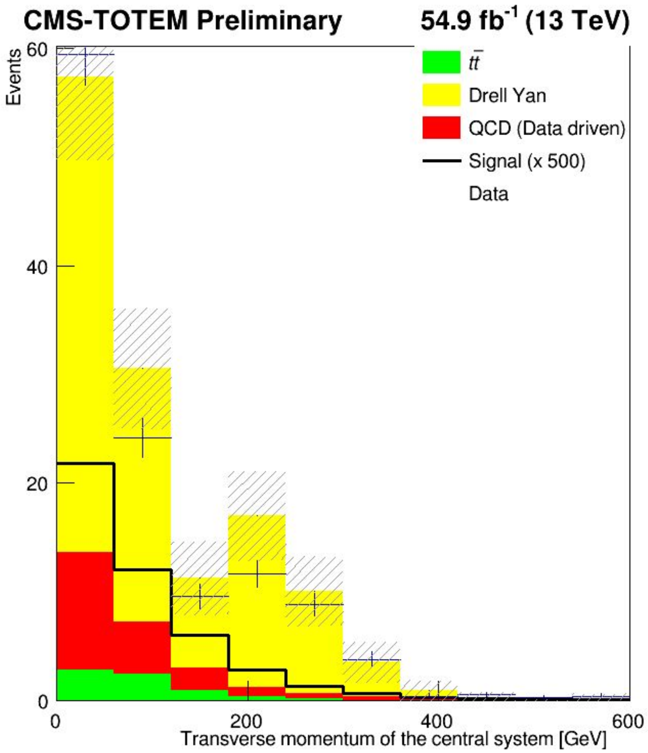




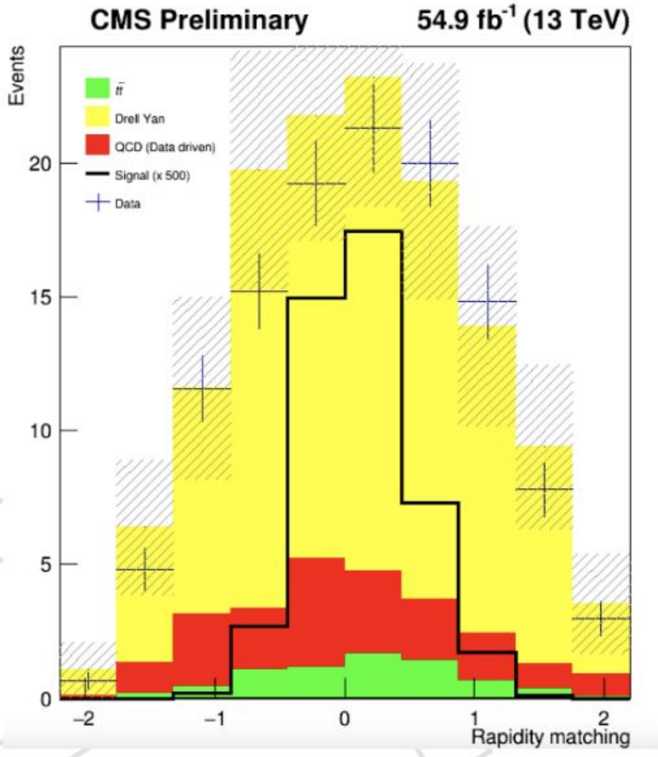
Transverse momentum of central system

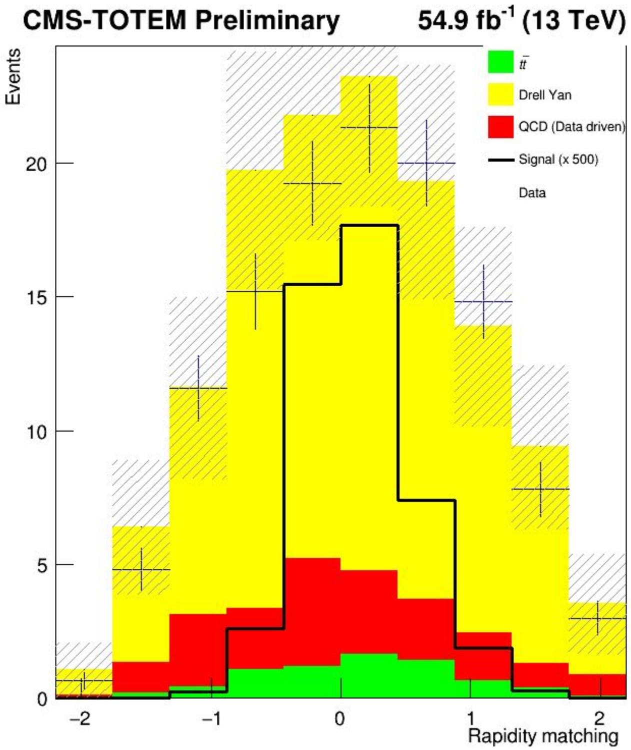
Original - left vs New - right





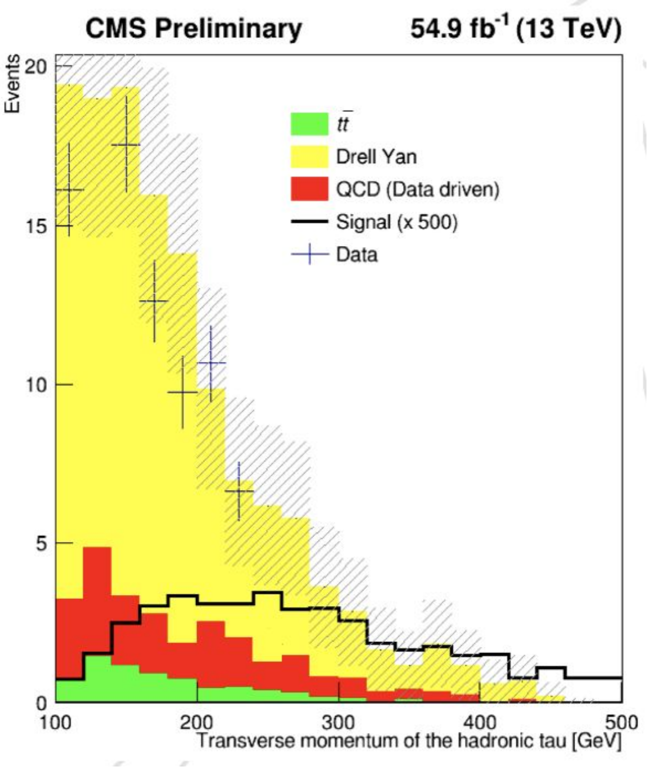
Rapidity matching
Original - left vs New - right

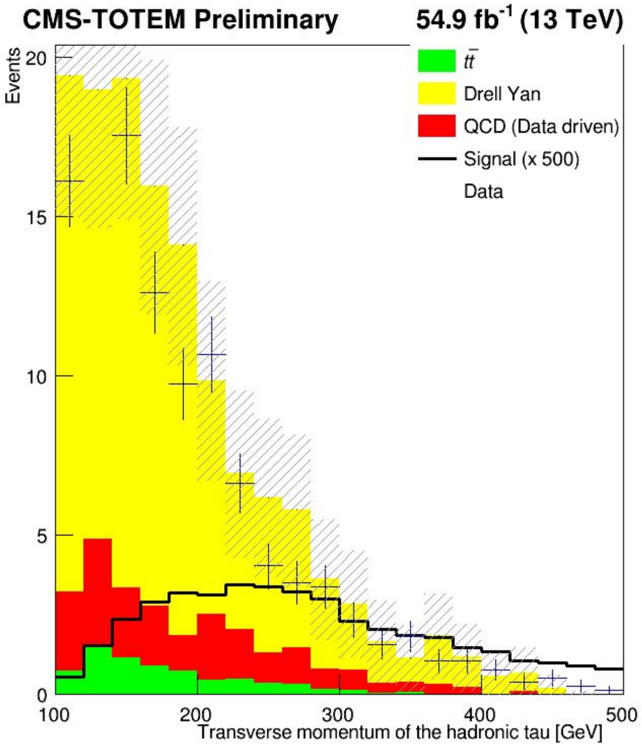




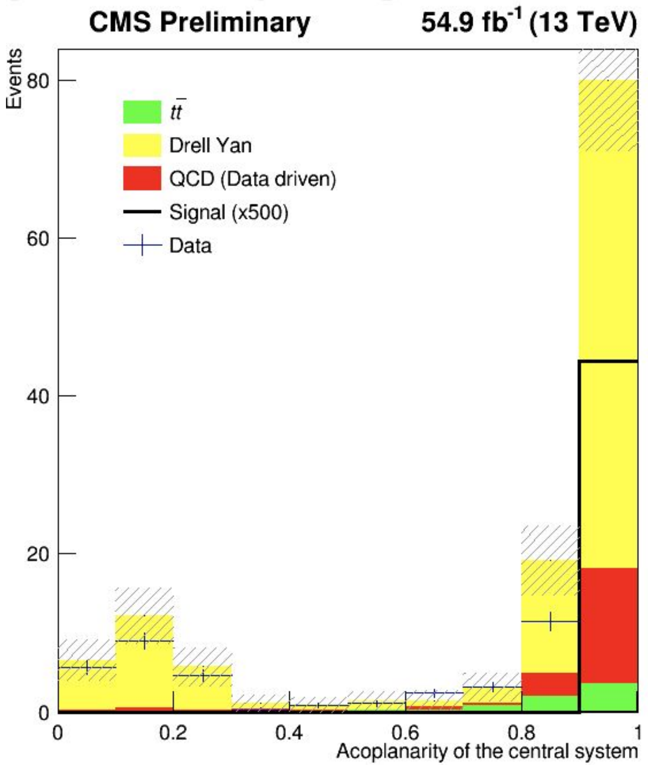
Transverse momentum of hadronic tau

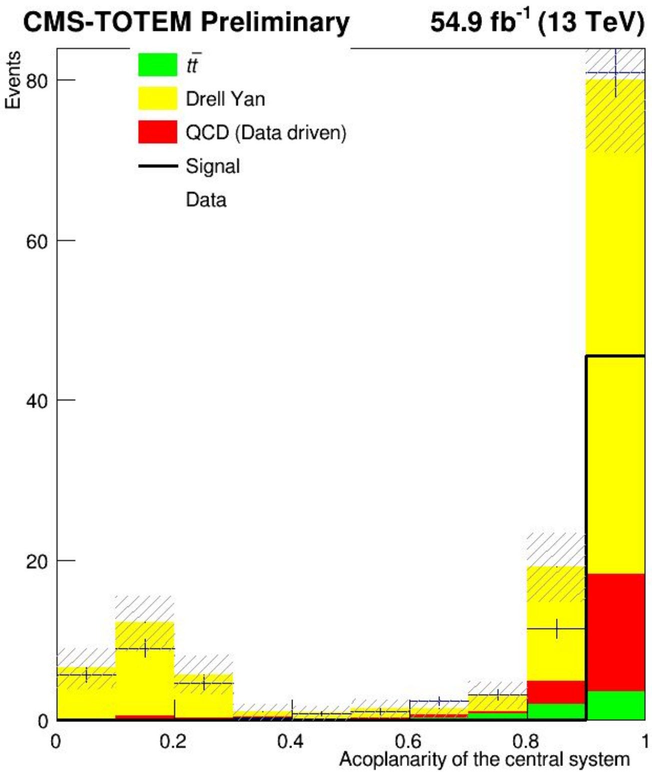
Original - left vs New - right





Acoplanarity of central system
Original - left vs New - right





Mystery weights

In original code

```
weight_sample = 54900.*(1.)/(4000.*1000.);
```

- Weight_sample is normalization factor
- weight_sample=0.013725

$$\text{weight\_sample} = \frac{\mathcal{L} \cdot \sigma}{N_{\text{gen}}}$$

What worked to make the plots the same:

```
weight_sample= 0.000013725;
```

Is original weight_sample/1000 -> UNITS

Mystery weights

```
//double w_data=1.;
double w_qcd=1.*0.8*0.13;
double w_ttjets=0.15*0.8*0.13;
double w_dy=1.81*0.8*0.13;
double w_sinal;
double w_data=1*0.13;
```

In 2018, due to an improved PPS detector configuration, silicon pixels are used. They allow the reconstruction of multiple protons and the scenario is different. Therefore, the technique used is the following:

- We estimate the probability $p_{n,m}$ to get n protons in arm 0 and m protons in arm 1;
- For the background samples, we observed that the probability to have at least one PU proton per arm is just 13%. Therefore, the weigth of each background event is rescaled of a factor 0.13;
- Subsequently, each background event is enriched with n protons in arm 0 and m protons in arm 1, according to the probability $p_{n,m}$ calculated earlier. If we get more than one proton per arm, the pair with the largest ξ is chosen;
- For the signal samples, they are directly enriched with the number of protons calculated according to the probability $p_{n,m}$.

```
double weight = .15; //w= Numero monte carlo / Numero de 2018 , Numero de 2018 = luminosidade de 2018 * secção de 2018
```

0.13 is due to pps reconstruction technique -> probability of having at least one proton per arm

Still no explanation for 0.8 factor on the background only

Background

Starts with data in NANOAOD format
DY and ttbar are MC
QCD is based on same sign events from data

↓

Fase0

QCD_fase0.cpp
DY_fase0.cpp
Ttjets_fase.cpp

↓

Produces fase0 files for each background used in:

↓

Fase1

nanotry_fase1_QCD.cpp
nanotry_fase1_DY.cpp
nanotry_fase1_jets.cpp

↓

Produces fase1, one file for each , after enriched with pileup protons

→ `QCD_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`
→ `DY_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`
→ `ttJets_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

Signal

Starts with data in MINIAOD format

↓

Sinal.cpp → data/MC corrections for the efficiency, scale, and resolution of taus, electrons, muons, and protons

↓

Produces ntuple with format similar as fase1, includes BSM weighs needed to normalize the sample for different SM and BSM scenarios

↓

`TauTau_sinal_PIC_july_2018.root`

final data files with the signal-like selection (opposite sign taus) :
`Dados_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

Used in plot_m.cpp
Produces normalized plots and the input distributions for BDT

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
  System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
  Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1)

Code organization
Background

Goal: only one file for signal and one file for background

(with parallelization using rootEnableImplicit)

File name	reads		Output
DY_fase0	`DY_2018_UL.txt`		`DY_2018_UL_skimmed_TauTau_nano_`
QCD_fase0	`QCD_2018_UL.txt` `dadosluminosidade.txt`	`Checks if luminosity in event matches with the data from file`	`QCD_2018_UL_skimmed_TauTau_nano_`
ttJets_fase0	`ttJets_2018_UL.txt`		`ttJets_2018_UL_skimmed_TauTau_nano_`
nanotry_fase1_DY	`DY_2018.txt` `Name of fase0 output files`		
nanotry_fase1_QCD	`QCD_fase1_PIC.txt` `Name of fase0 output files`		
nanotry_fase1_ttJets	`ttJets_fase1_PIC.txt` `Name of fase0 output files`		

Background-> fase0 specifics

Feature	`DY`	`ttjets`	`QCD`
Sample Type	Drell-Yan MC	$t\bar{t}$ MC	Data-driven QCD
Luminosity block filtering			✓
Extra gen matching	✓	✓	
Generator weight	1.81	0.15	
HLT version	`40_Trk1_eta2p1`	`40_Trk1_eta2p1`	`35_Trk1_eta2p1`

Code organization

Background-> fase1 specifics

Feature	`DY`	`ttjets`	`QCD`
Sample Type	Drell-Yan MC	tt̄ MC	Data-driven QCD
Scale Factors	Tau ID + energy scale + sys	Tau ID + energy scale	
Systematic Variations	full		
Charge Selection	Opposite Sign (`<0`)	Opposite Sign (`<0`)	Same Sign (`>0`)
Tau Selection Cuts	`pt > 100`, η `< 2.3`	`pt > 100`, η `< 2.4`	`pt > 100`, η `< 2.4`
Gen-Match Handling	yes		

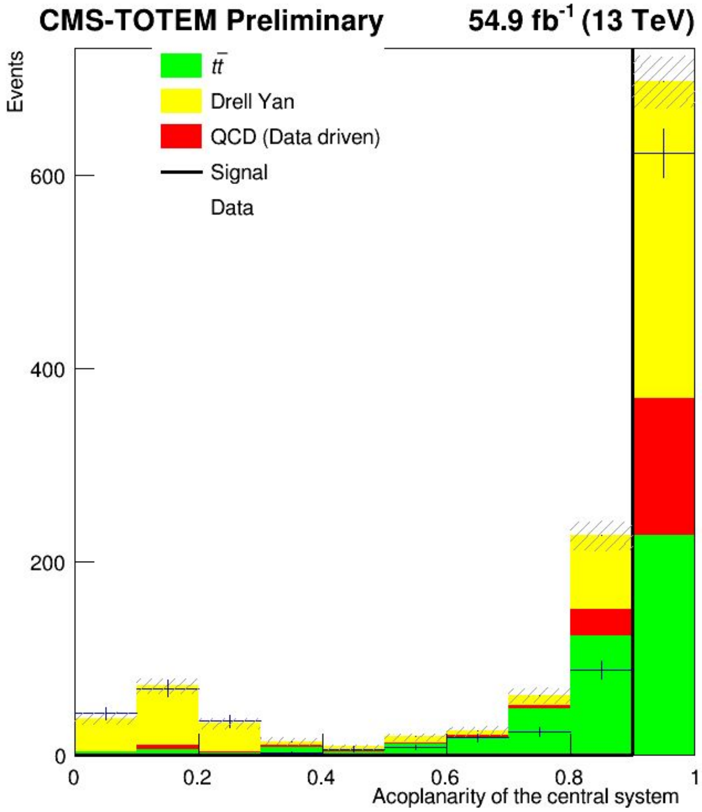
BDT: Code files information

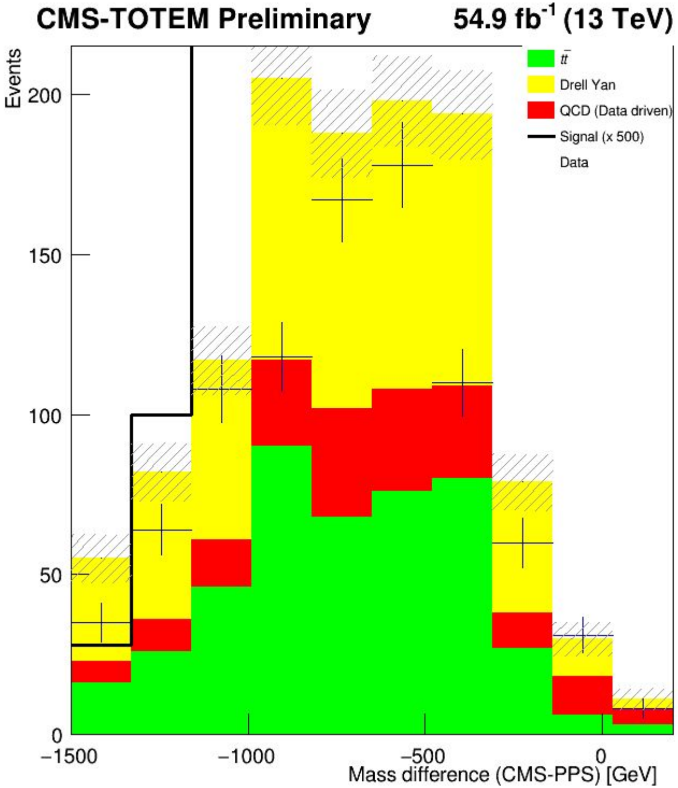
Variable	Description	Units
`sist_acop`	Central system acoplanarity	dimensionless
`sist_pt`	Total momentum	GeV
`sist_mass`	Invariant mass	GeV
`sist_rap`	System rapidity	dimensionless
`met_pt`	Missing energy	GeV
Mass matching	`sist_mass − sqrt(13000²×ξ₁×ξ₂)`	GeV
Rapidity matching	`sist_rap − 0.5×log(ξ₁/ξ₂)`	dimensionless

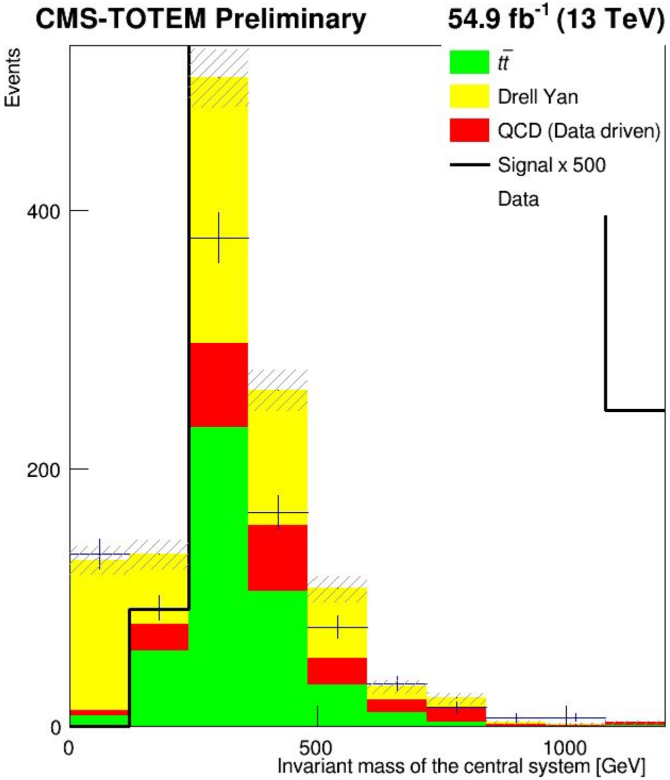
TMVAClassification	Train BDT/ML models	Background file Signal file	Output_TMVA_tautau_2018.root Dataset weights
TMVAClassificationApplication	Apply trained models to data	Trained models Data samples: dy, ttjets,qcd and signal	On file for each type containing BDT scores for every event
Bdt_output	Process BDT results	TMVApp_*.root Event weights, sample information	Analysis histograms Bdt score distributions signal/background separation cuts Cut optimization results
TMVAOutPut_Distributions	Create TMVA evaluation plots	Output_TMVA_tautau_2018.root Training results Test results Method performance data	ROC curves Variable importance plots Correlation matrices

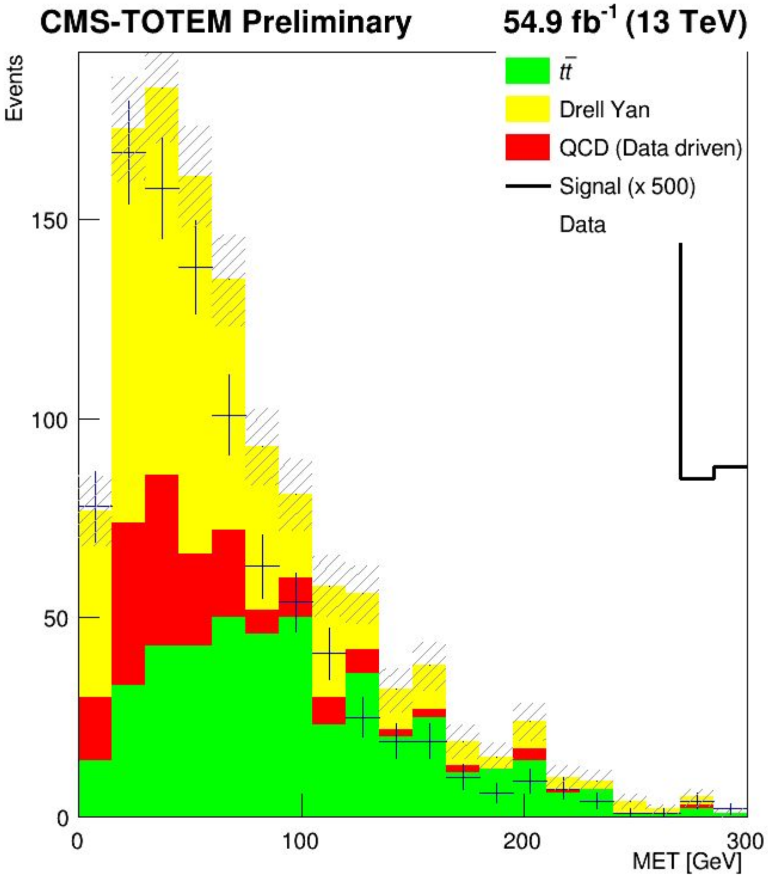
Plots of data without simulation weights applied

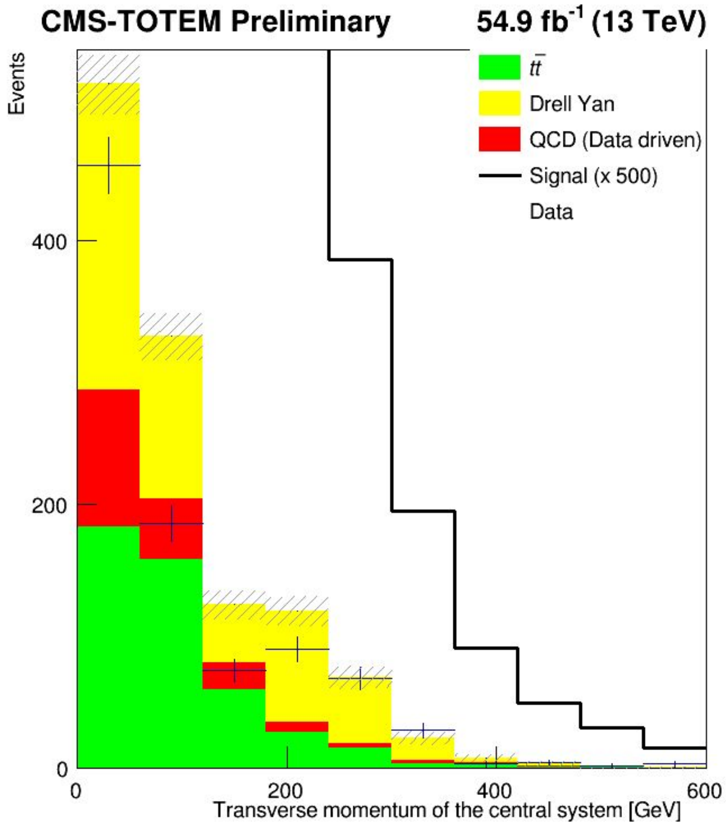
The following plots were obtained by setting the weight of each sample to 1

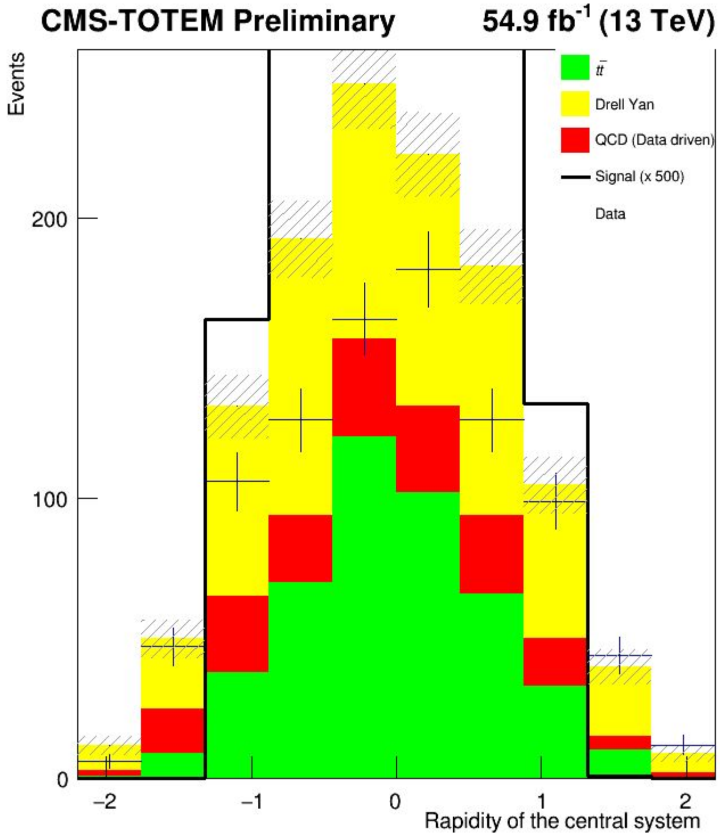

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal
Data
0
200
400
600
0
0.2
0.4
0.6
0.8
1
Acoplanarity of the central system

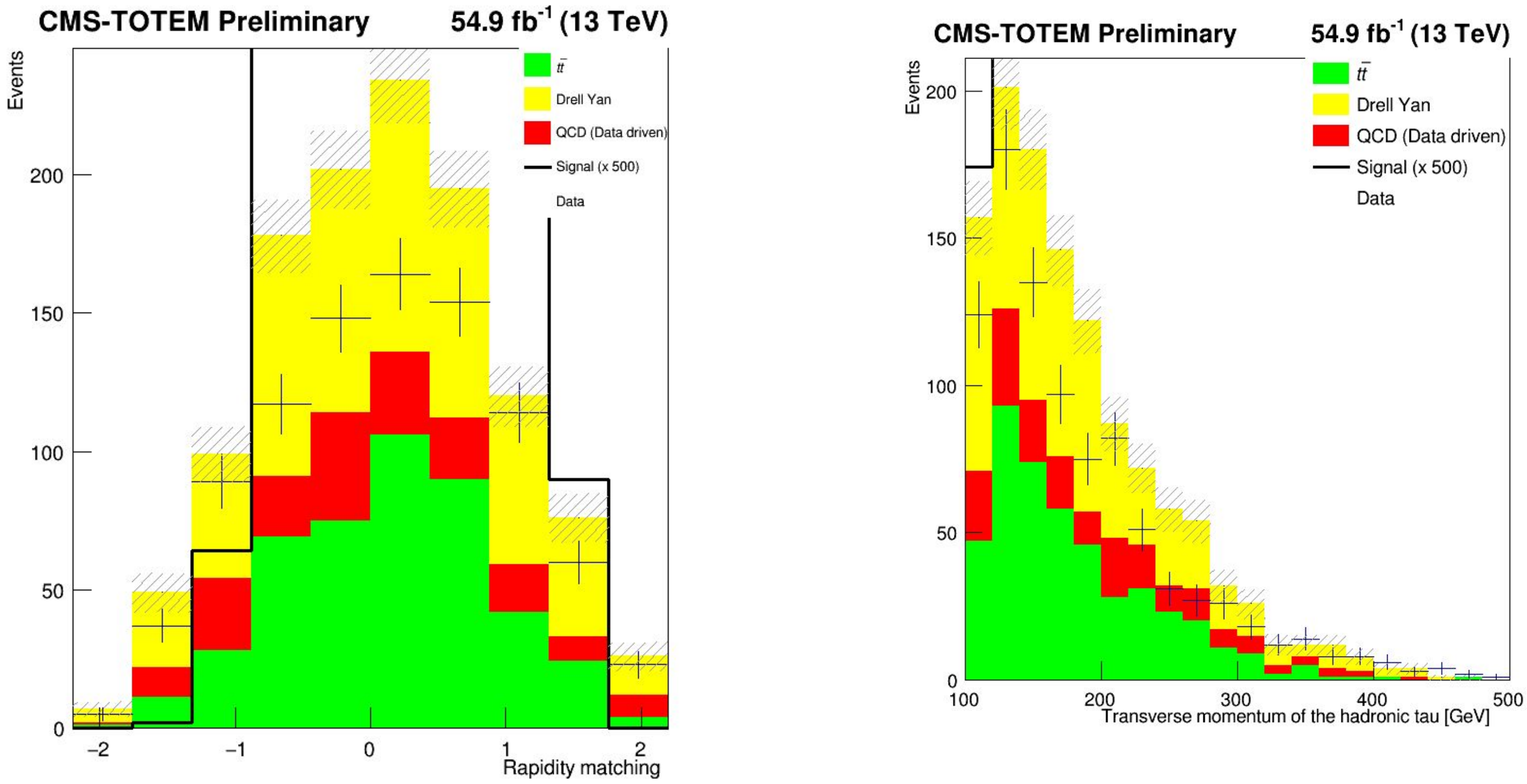

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
0
50
100
150
200
−1500
−1000
−500
0
Mass difference (CMS-PPS) [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal x 500
Data
400
200
0
0
500
1000
Invariant mass of the central system [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
150
100
50
0
0
100
200
300
MET [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
400
200
0
0
200
400
600
Transverse momentum of the central system [GeV]

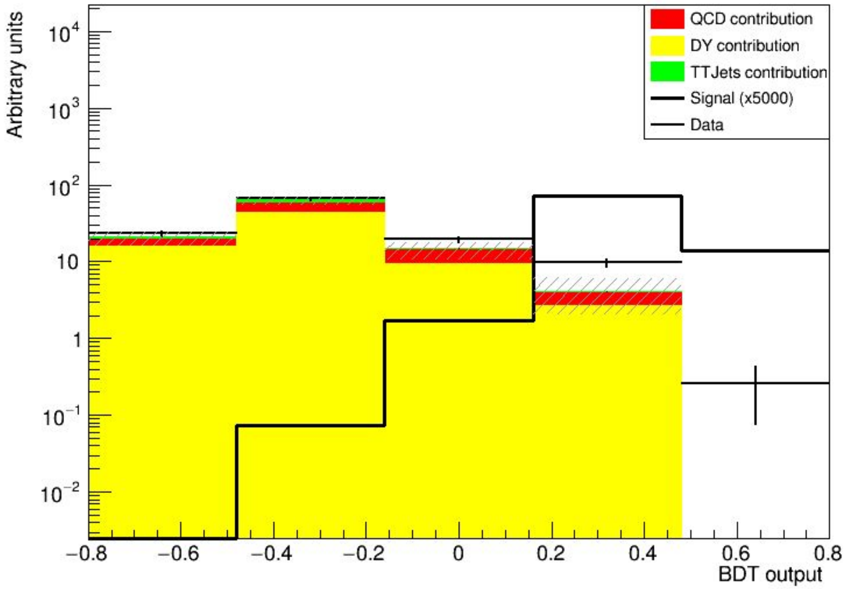

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
200
100
0
−2
−1
0
1
2
Rapidity of the central system


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Rapidity matching
Transverse momentum of the hadronic tau [GeV]

BDT output scores for training with previous histograms

Same shape as before changing the weights

-> weights only change the size of the histograms, not the shape.



In red: not applied

In green: applied

Control regions: τhτh

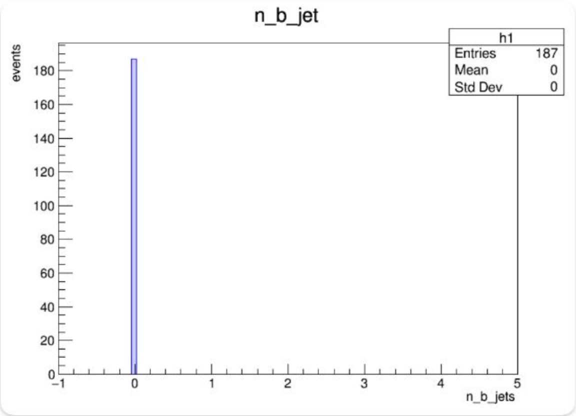
	Invariant mass (GeV)	acoplanarity	η	2 isolated ts	N of b-jets	p_t
DY	[40,100]	<0.3	<2.4	Opposite sign	0	
QCD	[100,300]	>0.8	<2.4	Same sign	0	<75
ttjets	[200,650]	>0.5	<2.4	Opposite sign	1 or more	<125

On top: original from analysis note
Bottom: new plots

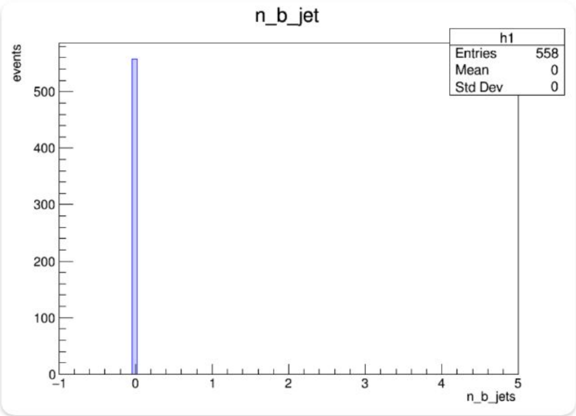
Number of b jets: 0 for all samples

Fase1

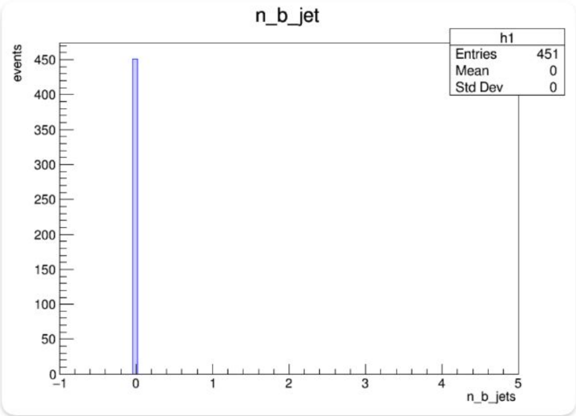
QCD



DY



ttJets

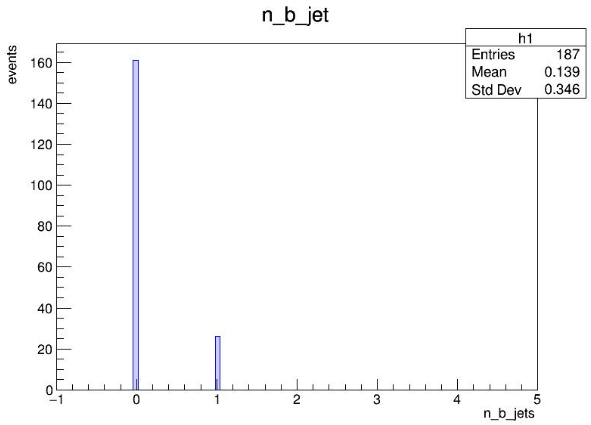


Cause: mismatch in the declaration of variable type in the proton enr

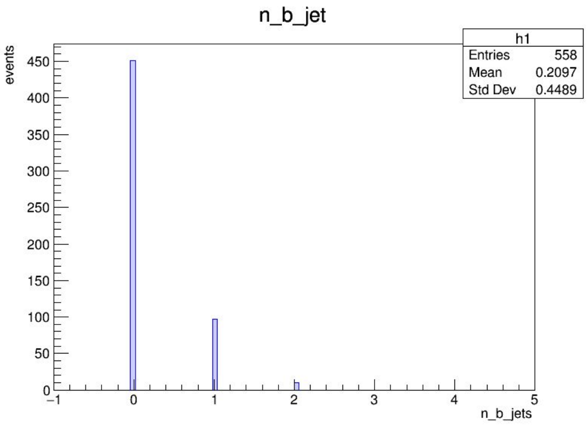
- Mismatch in the declaration of variable in the proton enrichment code (to introduce pileup protons);
    - N_b_jet was declared as *double* when it originally was *int*
- Handling of branch addresses

-> Fixed by passing the merged samples from phase1 through the proton enrichment code

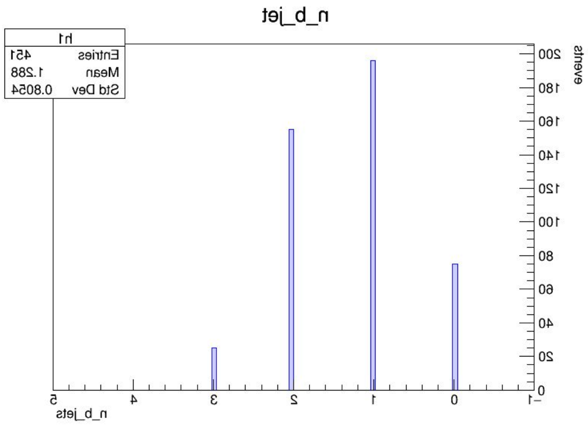
After fixing n_b_jet: Fase 1 samples



qcd

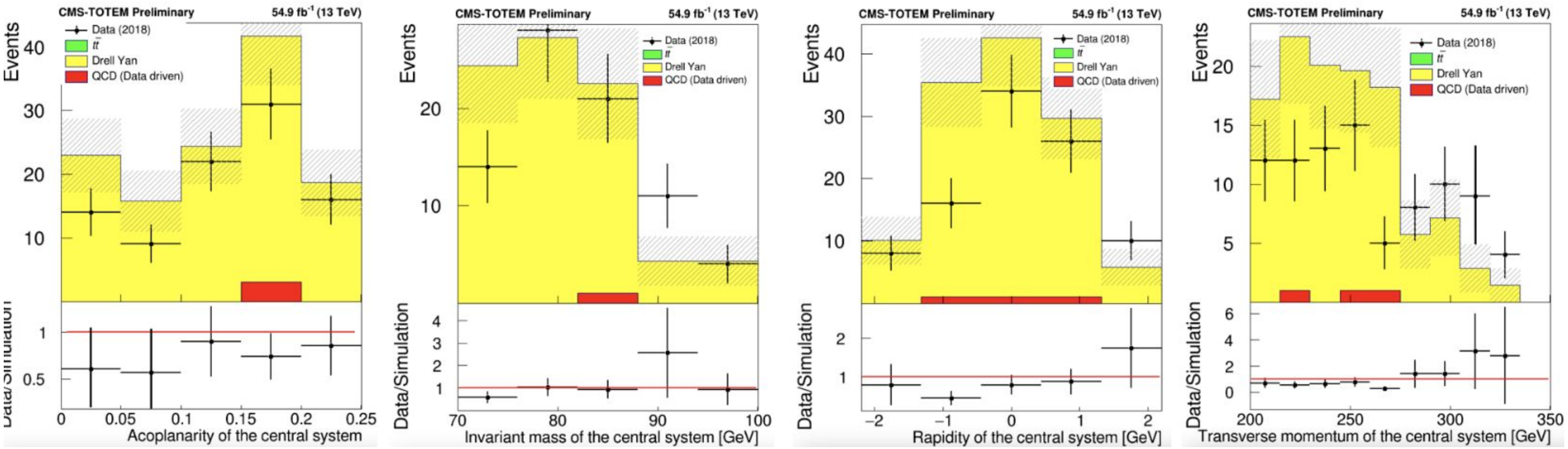


dy

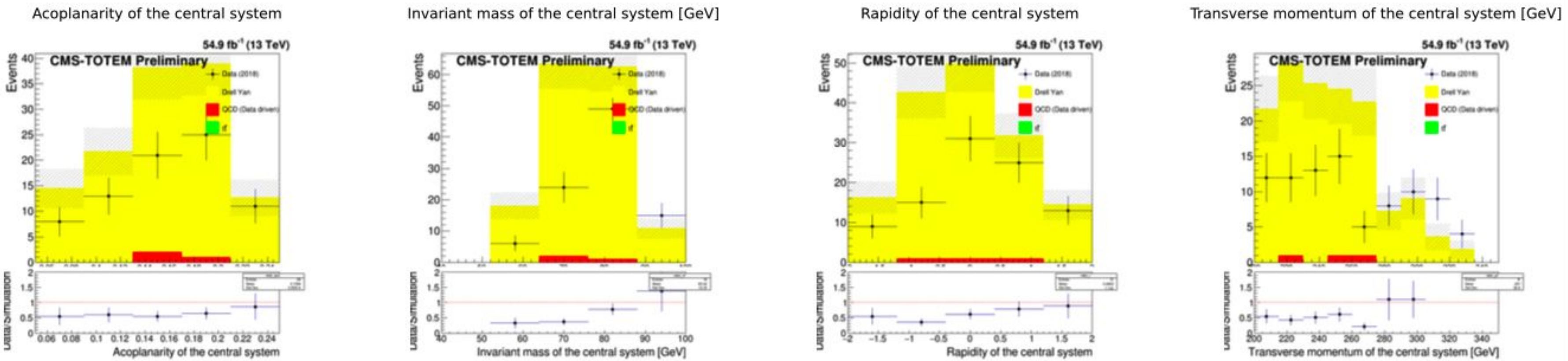


ttjets

DY control region: inv_mass [40,100]; acoplanarity < 0.3; n_b_jets=0 ;

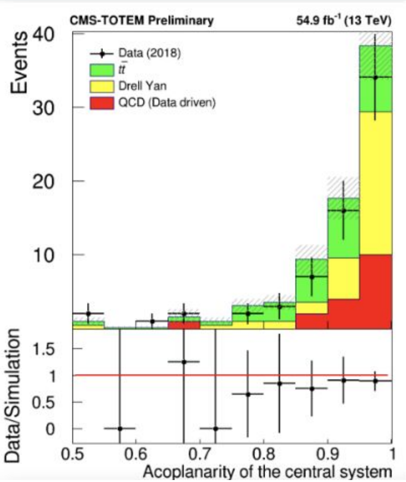


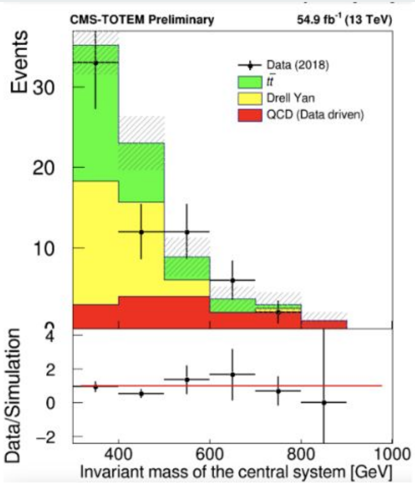
DY CR

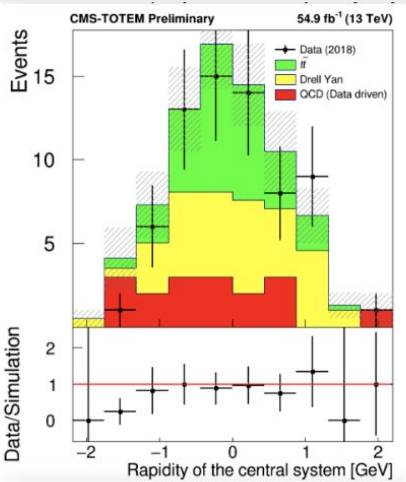


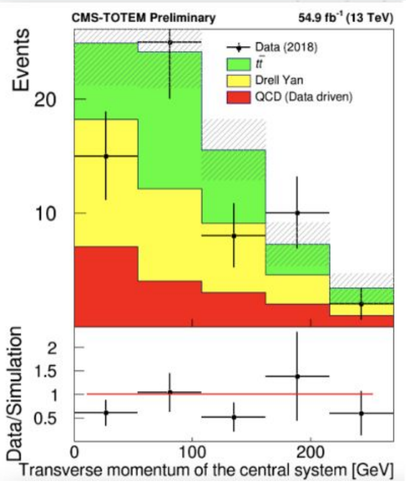
CR

TTJETS control region: inv_mass [200,650]; acoplanarity > 0.5; n_b_jets>=1 ; pt<125;

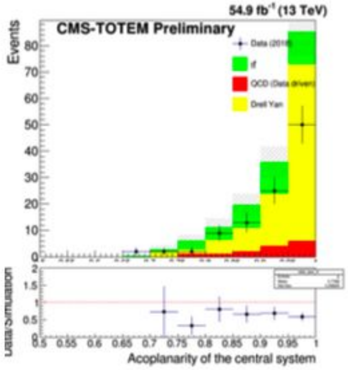


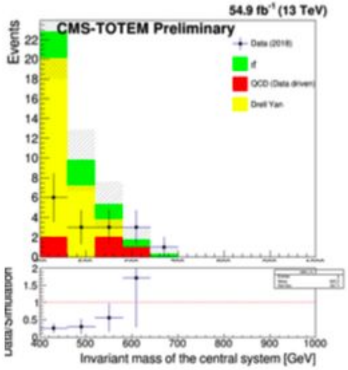


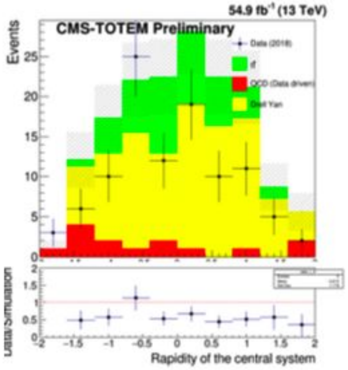


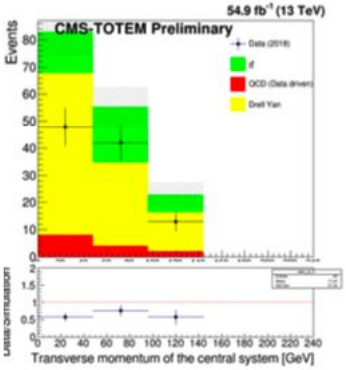


TT CR



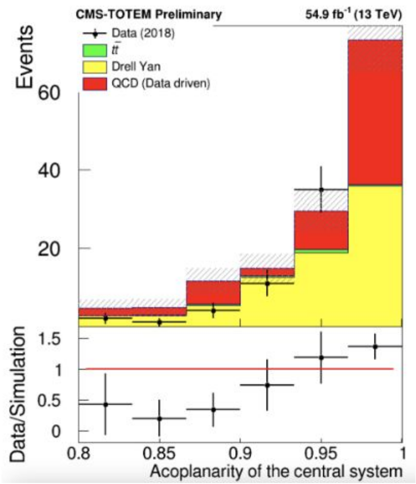


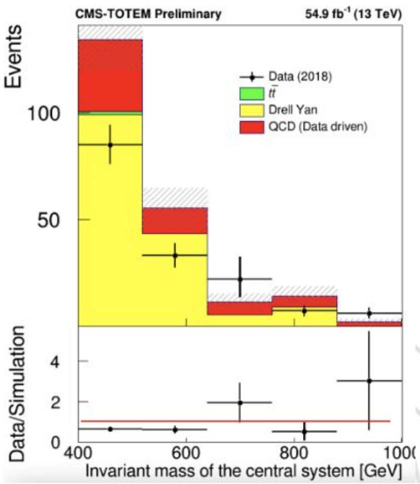


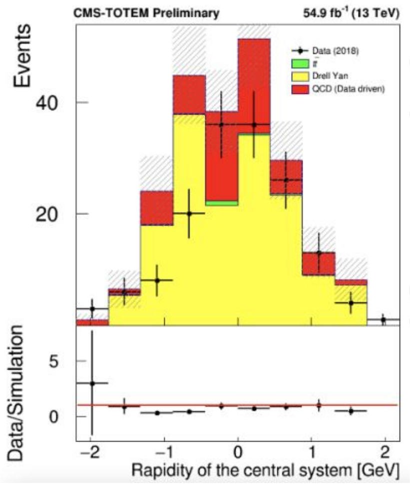


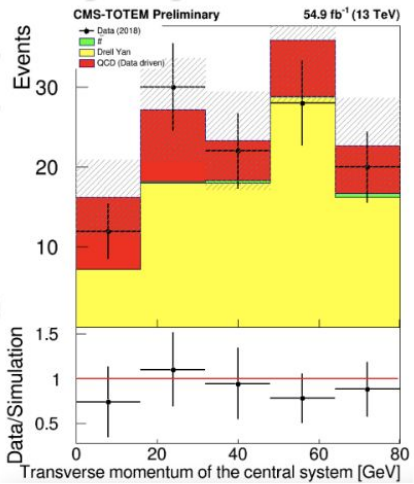
R

QCD control region: inv_mass [100,300]; acoplanarity < 0.3; n_b_jets<=0 ; pt<75;









QCD CR